



Artificial Intelligence for Planetary Health

Challenges, Opportunities, and Human Choice

The Climate Change AI Summer School 2026: Climate & Health Series

Sara Khalid *BE MSc DPhil Oxon*

Associate Professor | Centre for Statistics in Medicine

Head | Planetary Health Informatics Lab

University of Oxford

sara.khalid@ndorms.ox.ac.uk

www.csm.ox.ac.uk | [@PHI_Oxford](https://twitter.com/PHI_Oxford)

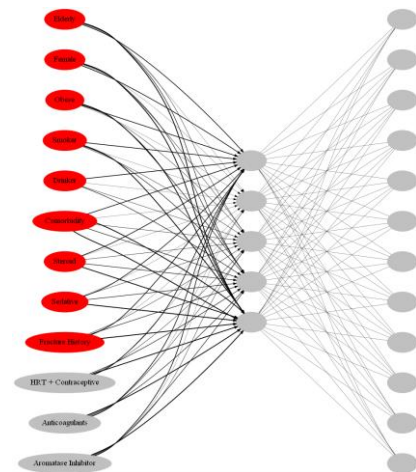
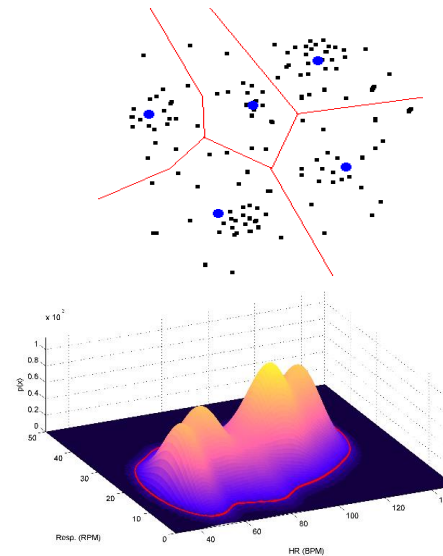
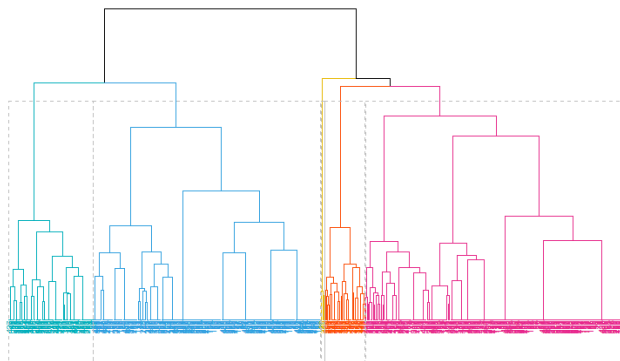
www.ndorms.ox.ac.uk/planetary-health-informatics



Planetary
Health
Informatics

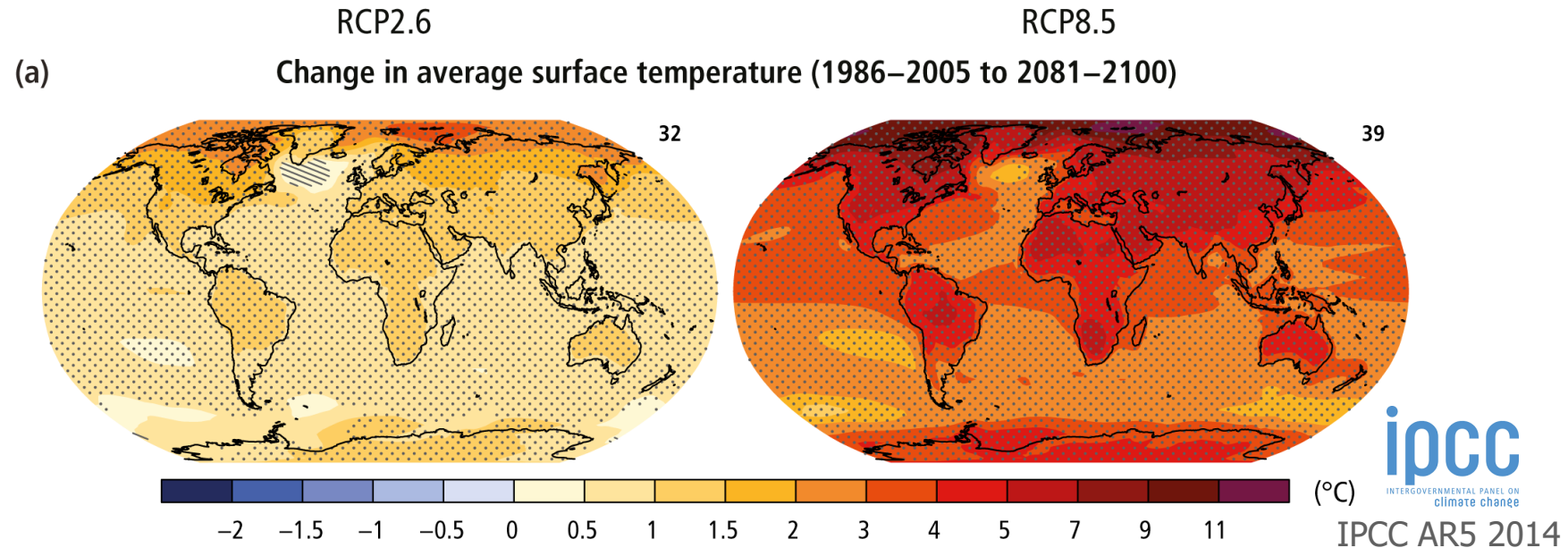
Outline

1. Climate Change and Planetary Health
2. The Role of Health Informatics
3. Challenges and Opportunities in Planetary Health Analytics

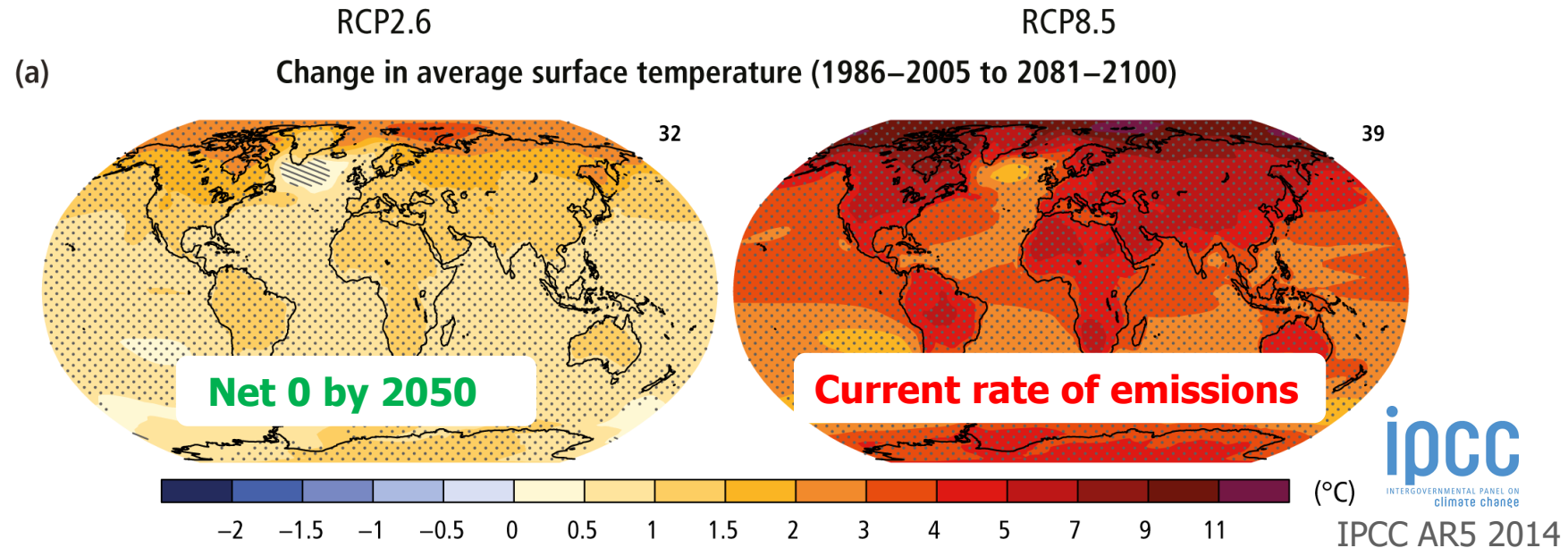


Climate Change & Planetary Health

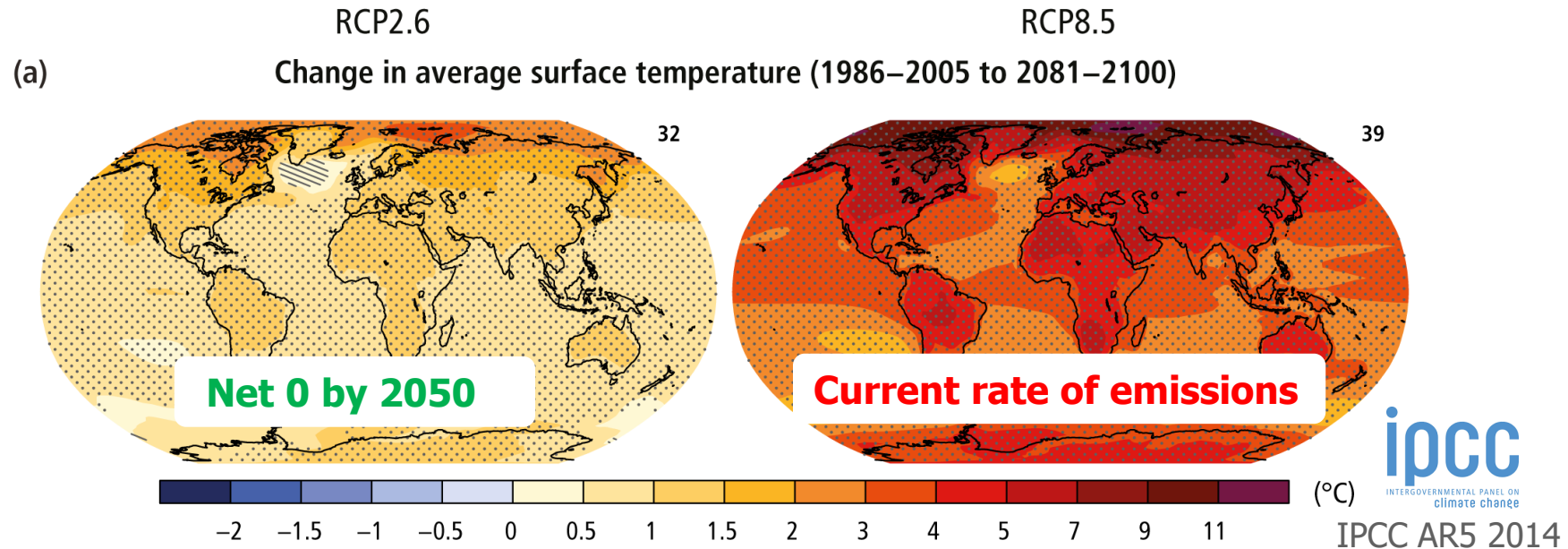
A Rapidly Warming World



A Rapidly Warming World



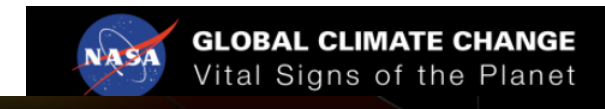
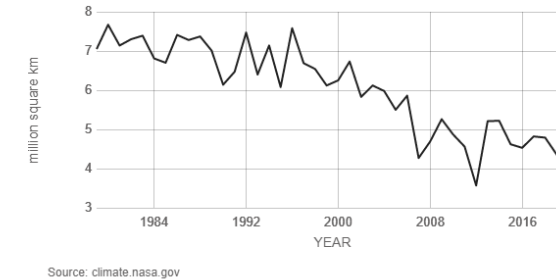
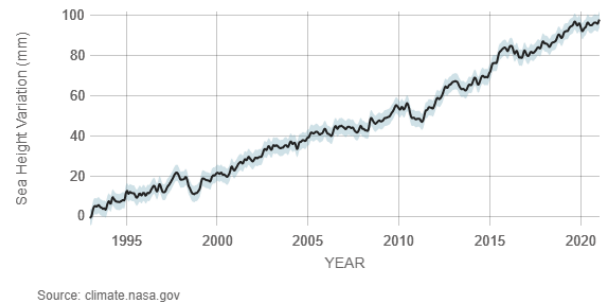
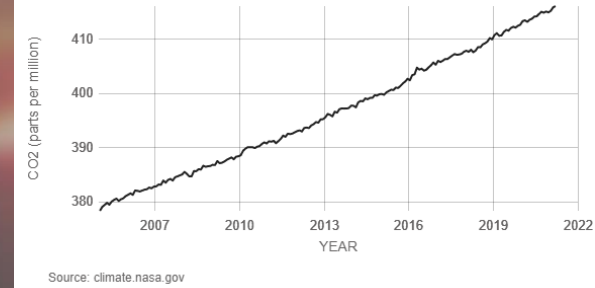
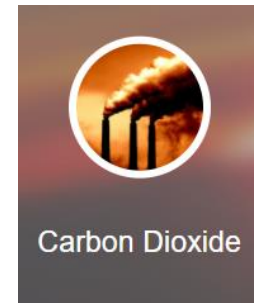
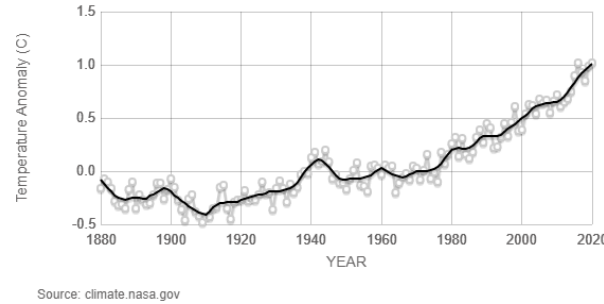
A Rapidly Warming World



“Climate change is the biggest global health threat of the 21st century”
Lancet Commission 2009

“Climate change threatens the very foundations of human health and wellbeing”
Lancet Countdown Report 2020

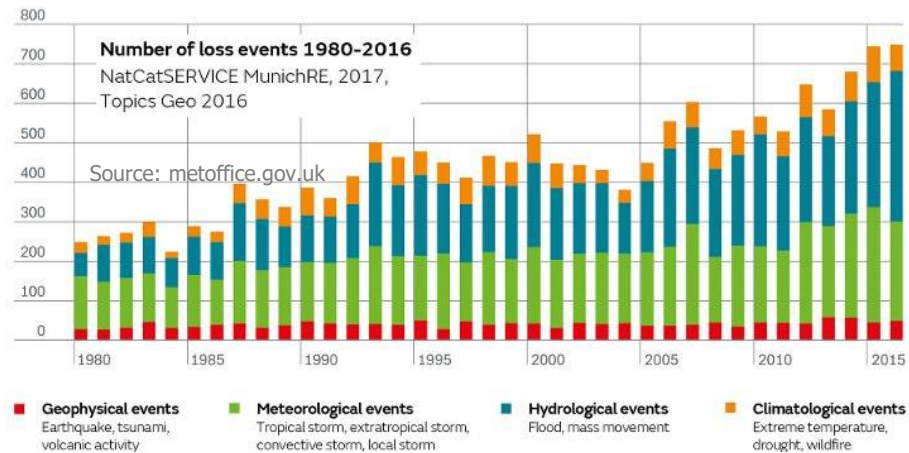
The Earth's Vital Signs



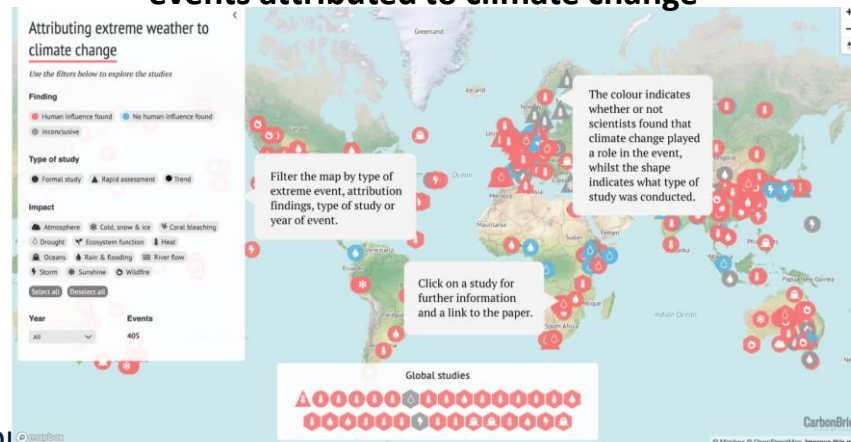
CARBON DIOXIDE	GLOBAL TEMPERATURE	ARCTIC ICE MINIMUM	ICE SHEETS	SEA LEVEL
↑ 416 parts per million	↑ 2.1 °F since 1880	↓ 13.1 percent per decade	↓ 429 billion metric tons per year	↑ 3.3 millimeters per year

Change, Variability, Weather

Are extremes becoming more frequent?

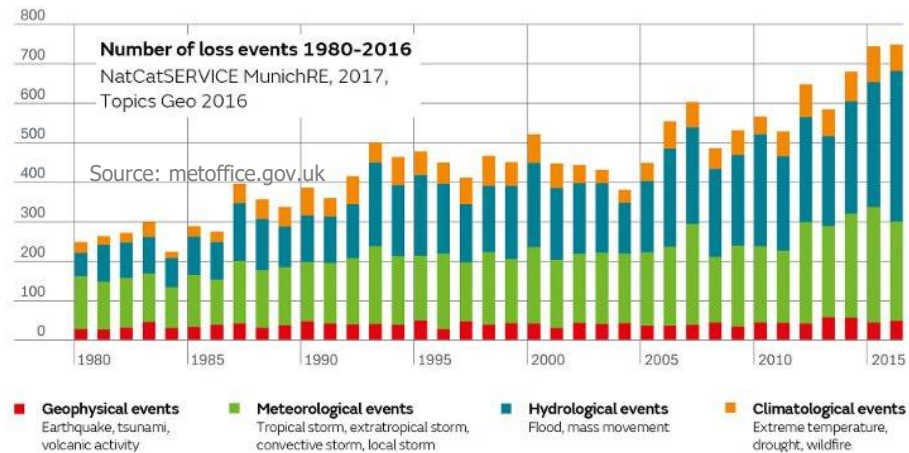


350+ peer-reviewed studies on 450 extreme weather events attributed to climate change

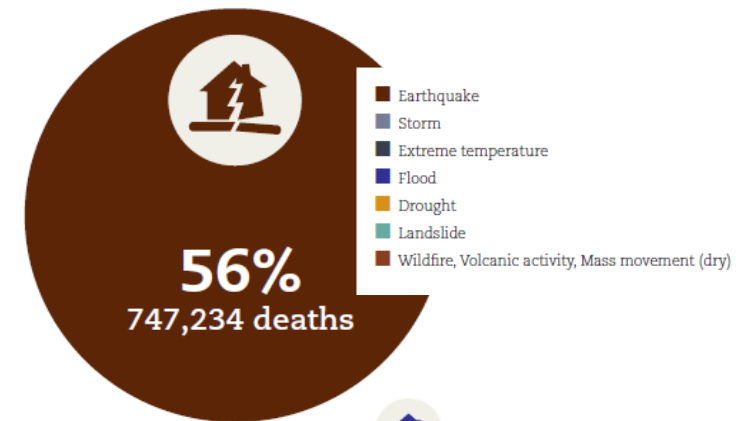


Change, Variability, Weather

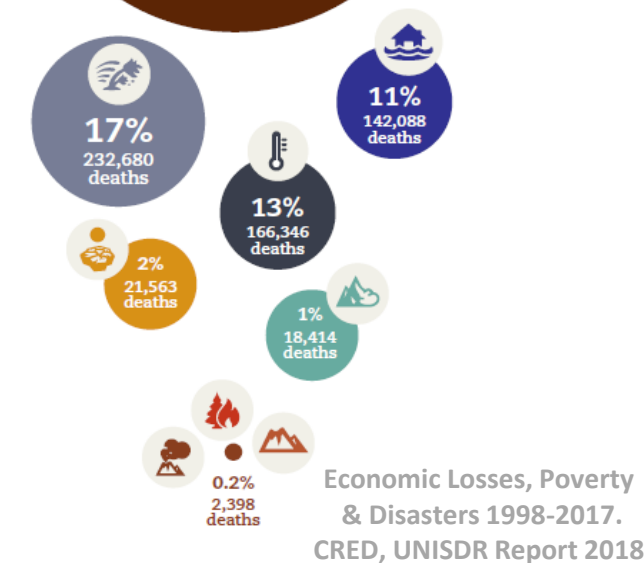
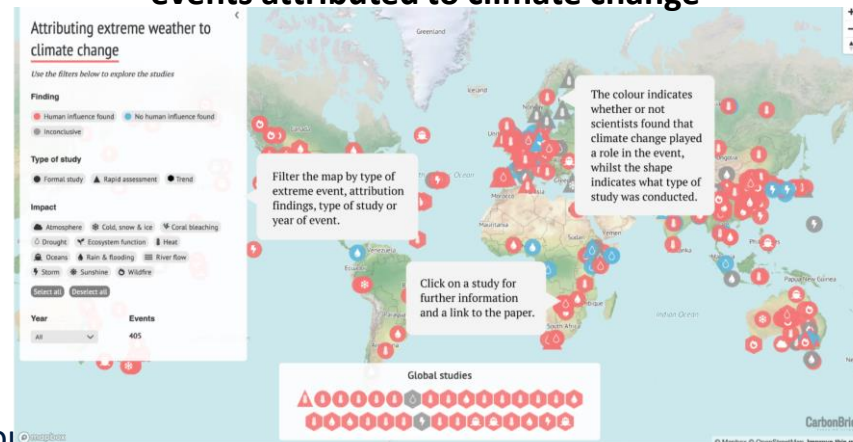
Are extremes becoming more frequent?



Number of deaths
per disaster type 1998-2017



350+ peer-reviewed studies on 450 extreme weather events attributed to climate change



Biodiversity Loss

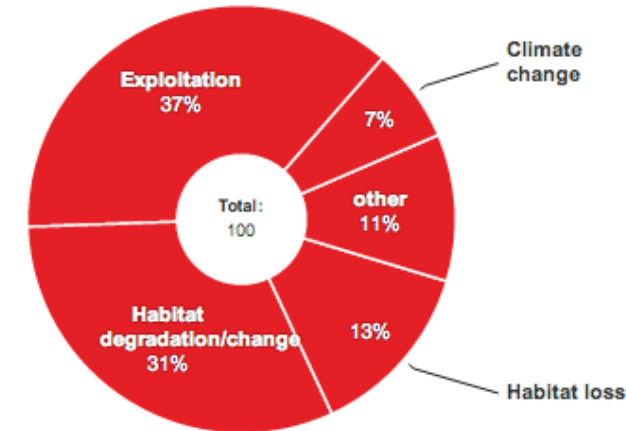
The
Guardian

Earth has lost half of its wildlife in the past 40 years, says WWF



Causes of wildlife declines globally

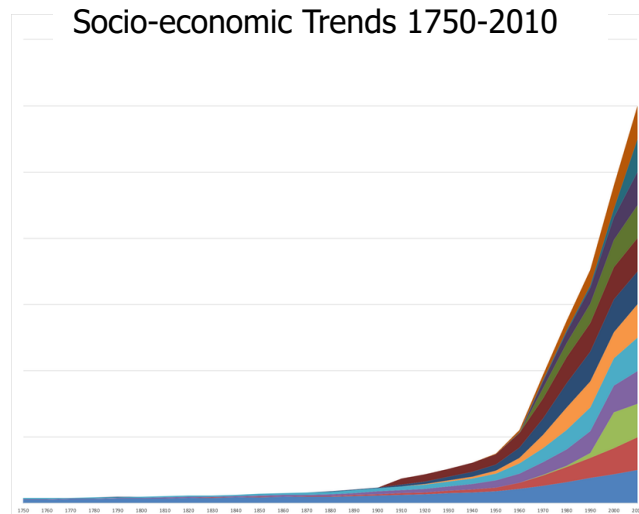
WWF's 'Living Planet Index' categorises the causes of wildlife declines based on analysis of 3,430 species' populations



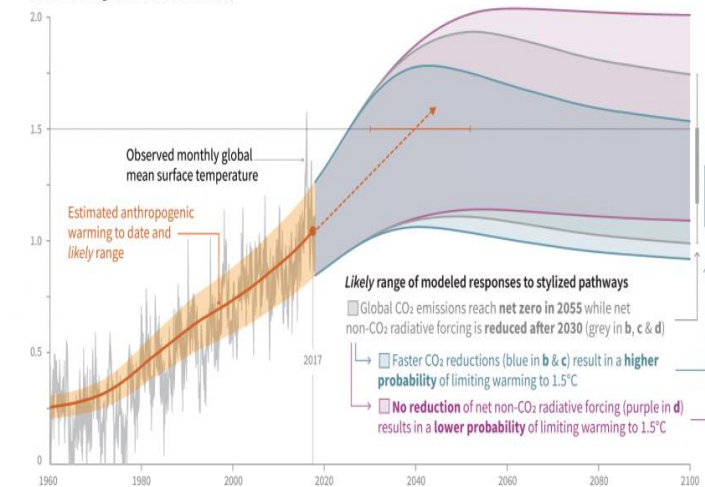
SOURCE: WWF

theguardian

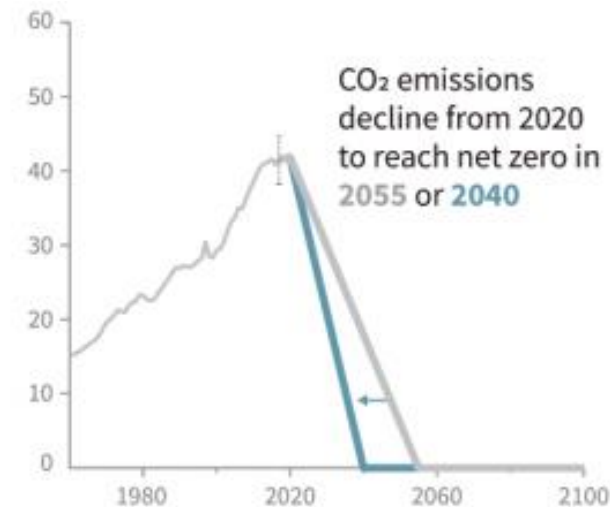
Acceleration and Tipping Point



Global Warming Relative to 1850 -1900

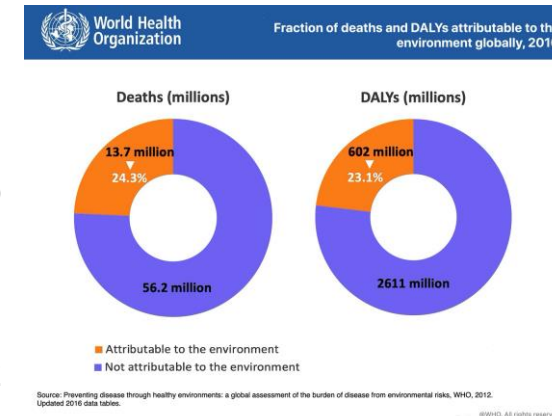


ipcc
INTERGOVERNMENTAL PANEL ON
climate change

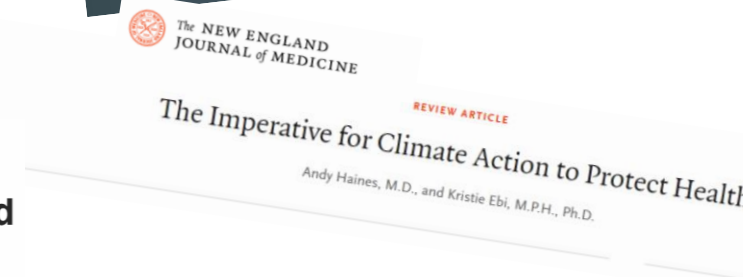


Links with Health

- “85 of 102 investigated diseases are related to the environment.”
 - WHO Preventing Disease Through Healthy Environment 2006
- 1/4 deaths worldwide linked to environment
 - WHO Preventing Disease Through Healthy Environment 2012

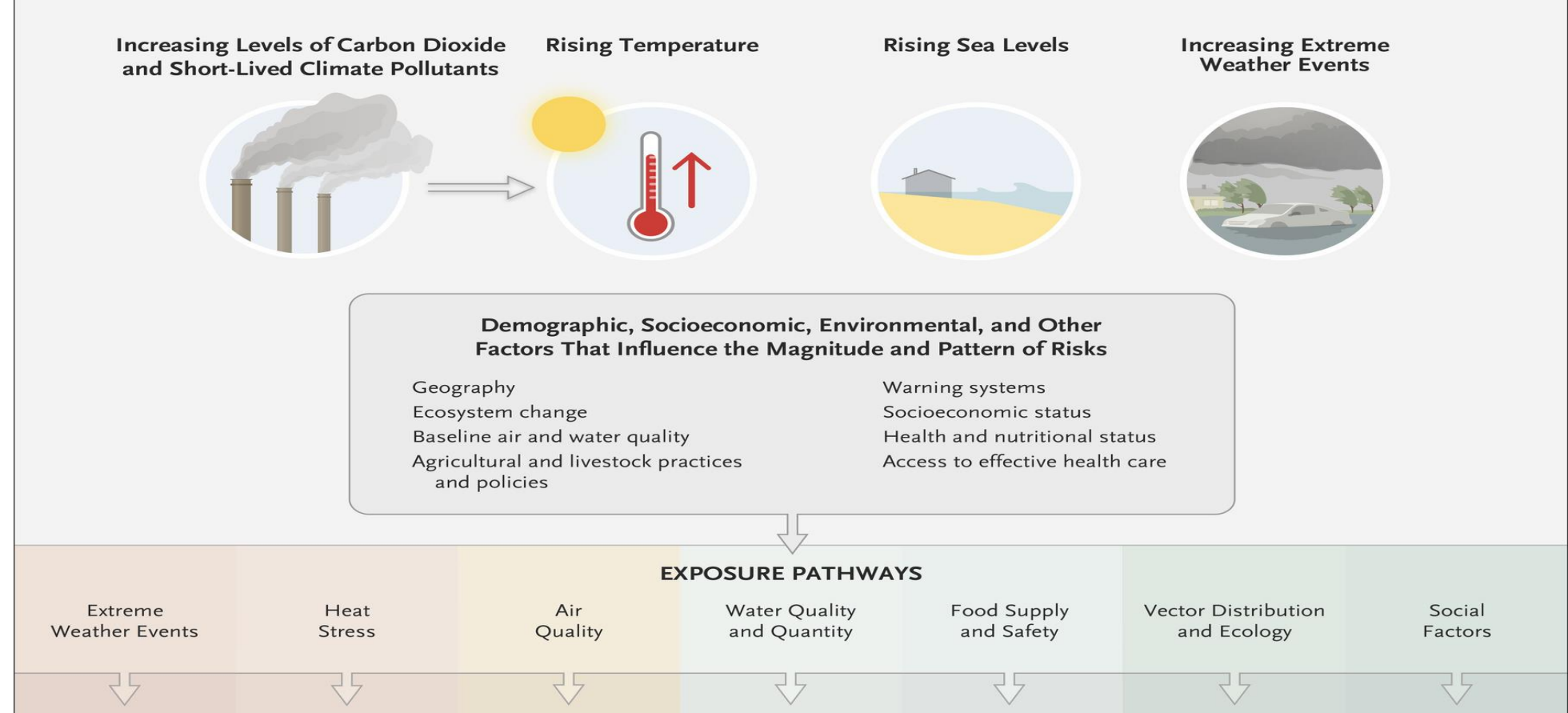


Deadly flooding. 'Zombie fires.' A massive iceberg. It's been a wild week in weather and climate change



Health Impacts

- Direct
- Indirect
- Long-term



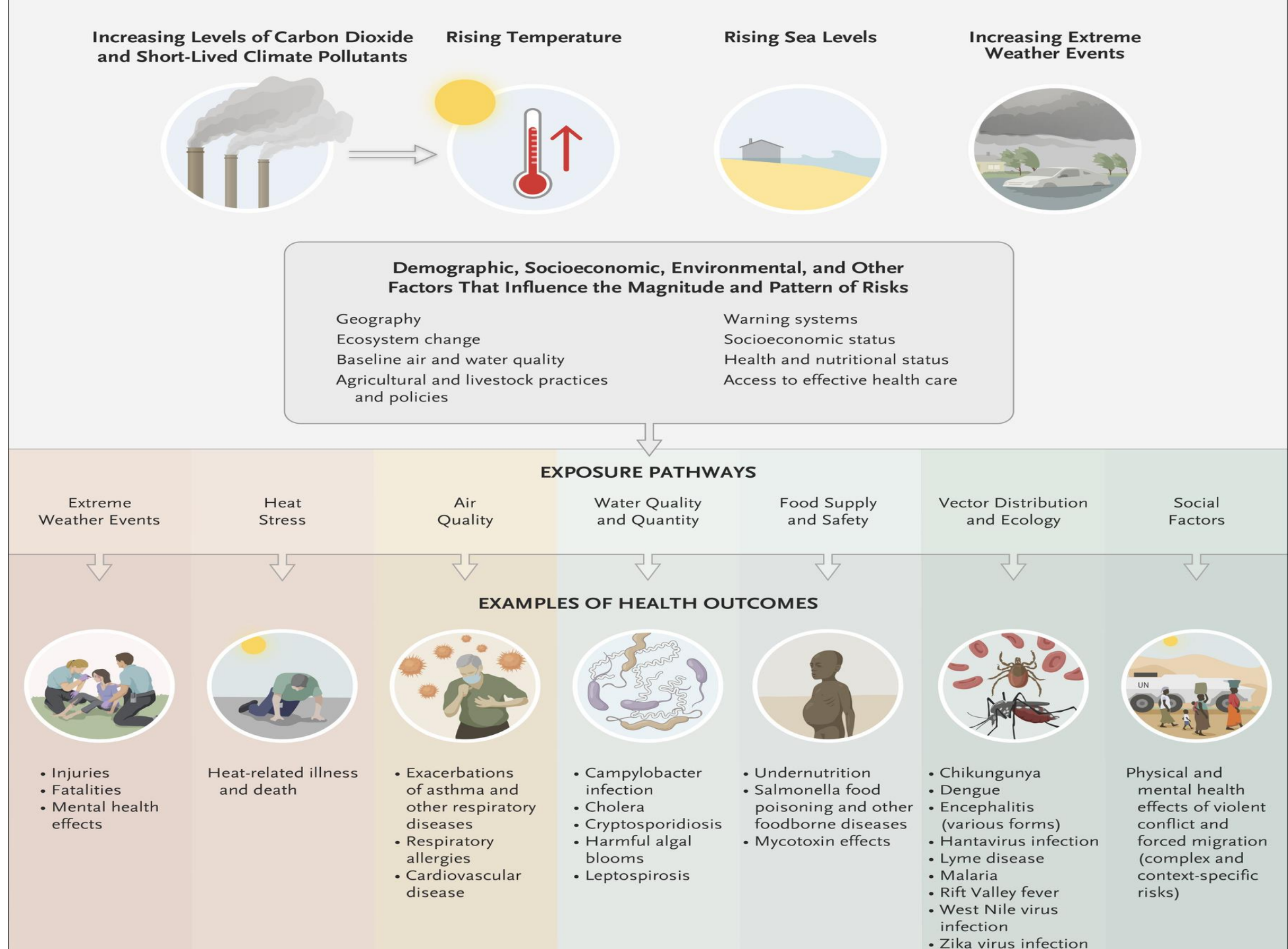
Haines et al. Engl J Med 2019

DOI:

10.1056/NEJMr1807873

Health Impacts

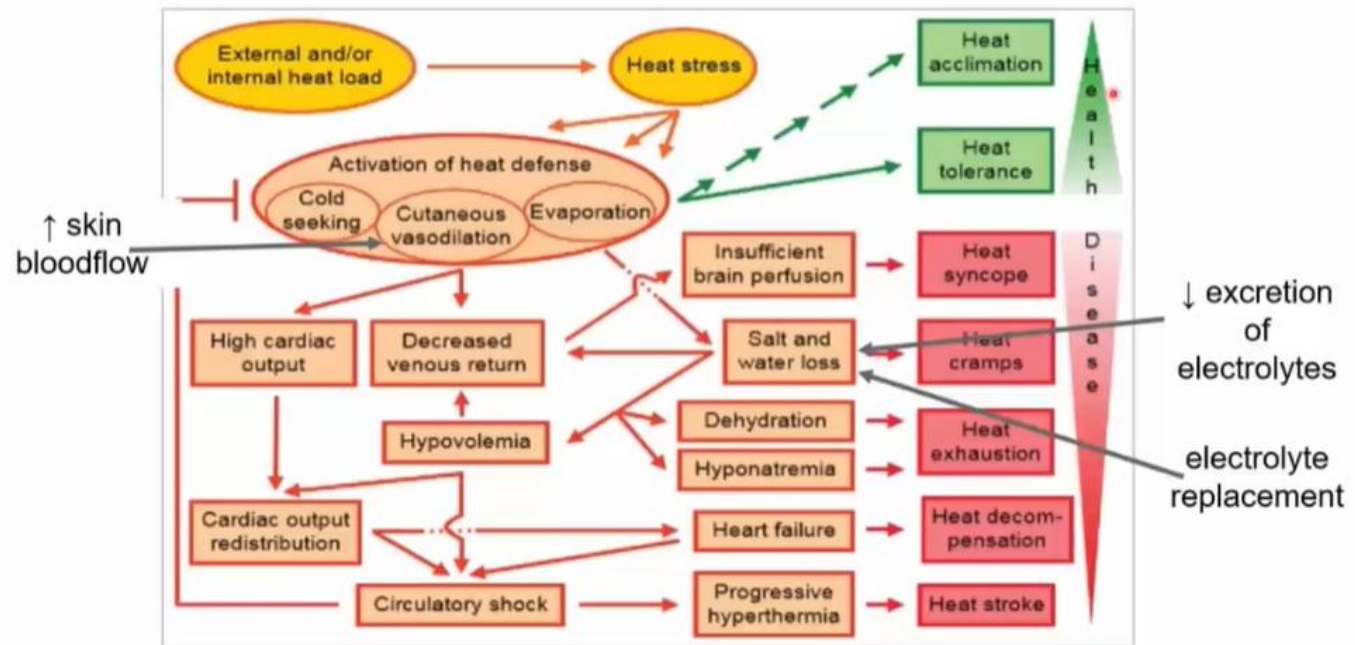
- Direct
- Indirect
- Long-term



Heat and Health

Mechanisms of Heat-Associated Illness

- Direct
 - Heat stroke
 - Cramps
 - Exhaustion
 - Hyper/hypo-thermia
- Chronic Conditions
 - Cardiovascular disease
 - Diabetes
 - Respiratory disease
 - Cerebrovascular disease
- Hospitalisations
 - Cardiovascular
 - renal
 - respiratory disorders



Szekely M et al. PMID: 27227063

Heat and Health

- **Who is most vulnerable?**

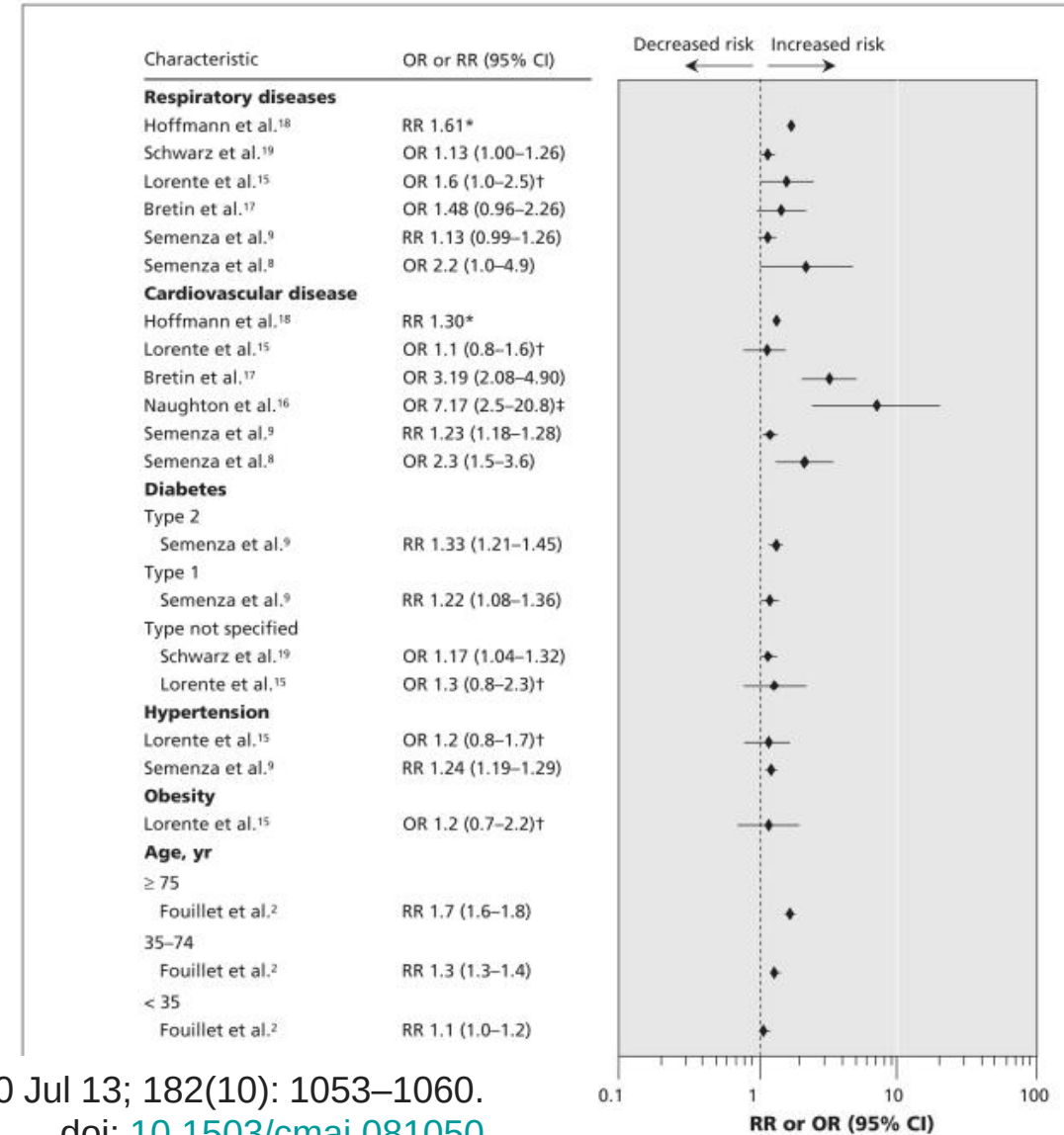
Heat and Health

- **Who is most vulnerable?**

- Multi-morbidity, poly-pharmacy
- Older adults
- Young children
- Pregnancy, menopause in women
- Outdoor, unventilated occupations

- **Social determinants and Disparities**

- Socioeconomic status
- Lifestyle factors
- Ethnicity/Genetics



[Kenny et al. CMAJ. 2010 Jul 13; 182\(10\): 1053–1060.](#)

doi: [10.1503/cmaj.081050](https://doi.org/10.1503/cmaj.081050)

Heat and Health

- Most Vulnerable
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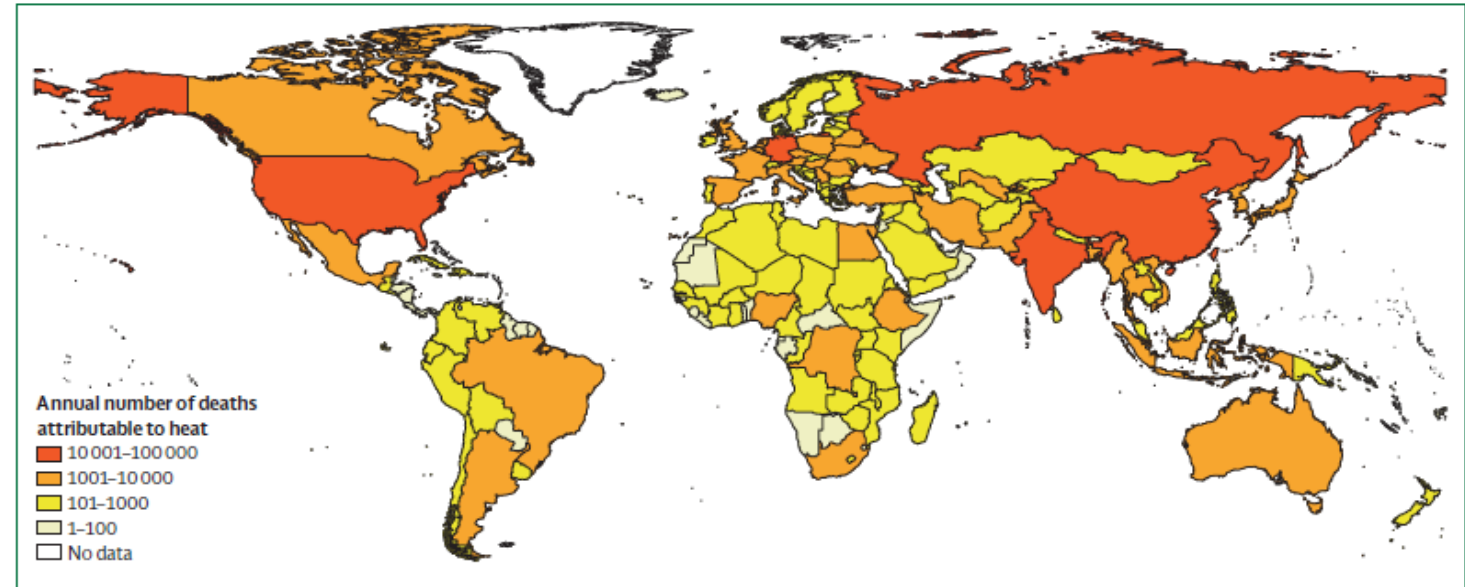


Figure 3: Annual heat-related mortality in the population older than 65 years averaged from 2014 to 2018

“Over 50% increase in heat-related mortality in people older than 65 years, resulting in a total of 296 000 deaths in 2018.”

Lancet Countdown 2015

Air Pollution and Mortality

Household air pollution

3.8 million

deaths every year as a result of exposure to indoor
smoke from cooking fuels

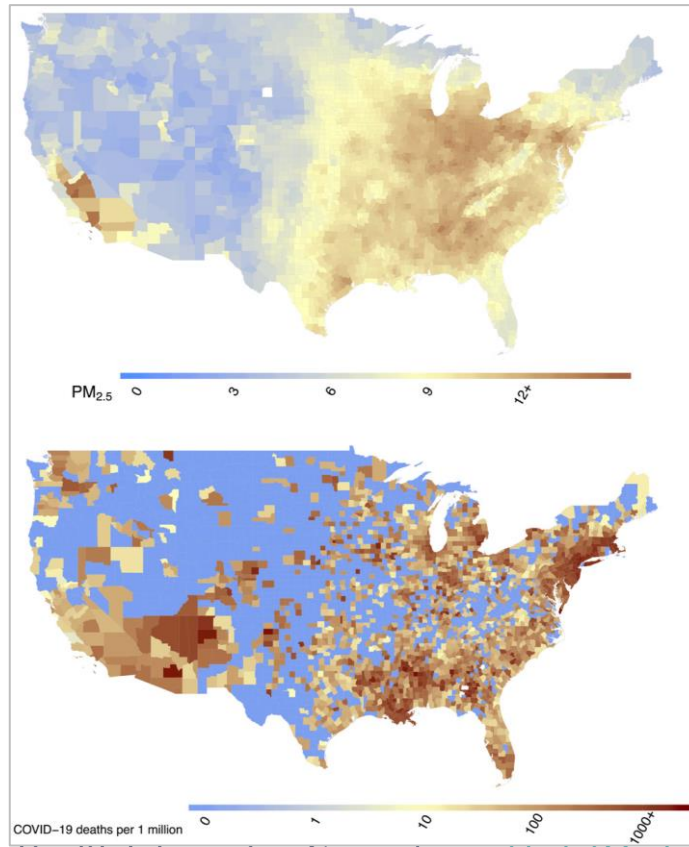
4.2 million

deaths every year as a result of exposure to fine
particulate matter

www.who.int/data

Air Pollution and Mortality

Historical PM_{2.5} concentrations (top)
and COVID-19 deaths (bottom) in
the US.



Household air pollution

3.8 million

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smoke from cooking fuels

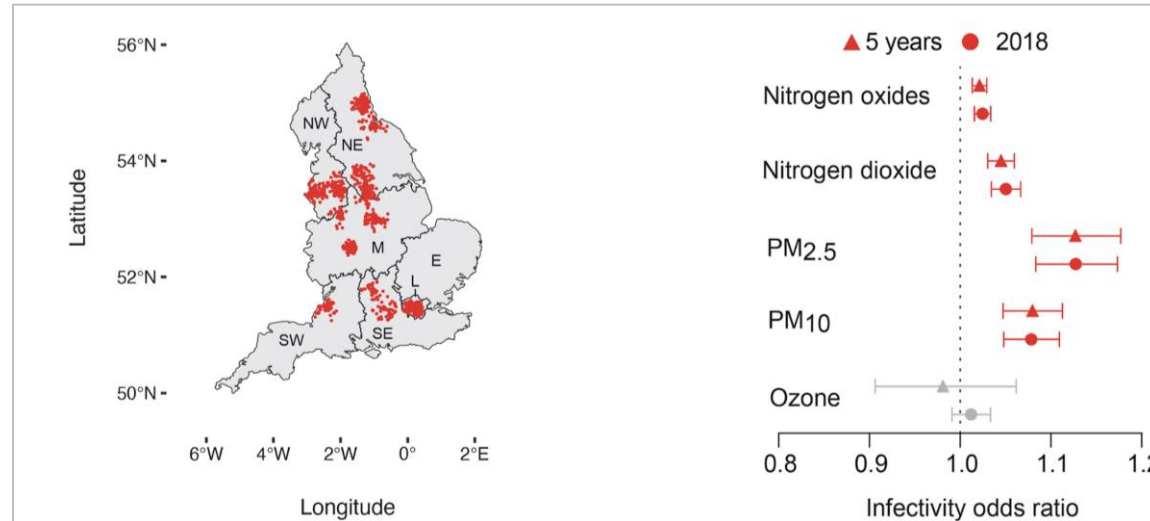
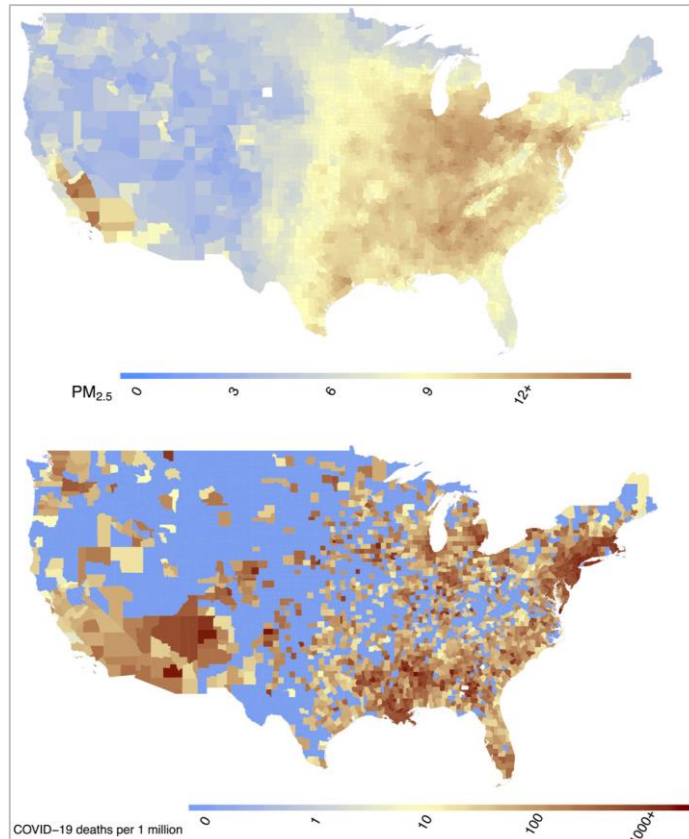
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Air Pollution and Mortality

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Association between exposure to air pollutants and the number of lab-confirmed COVID-19 cases in the UK Biobank.

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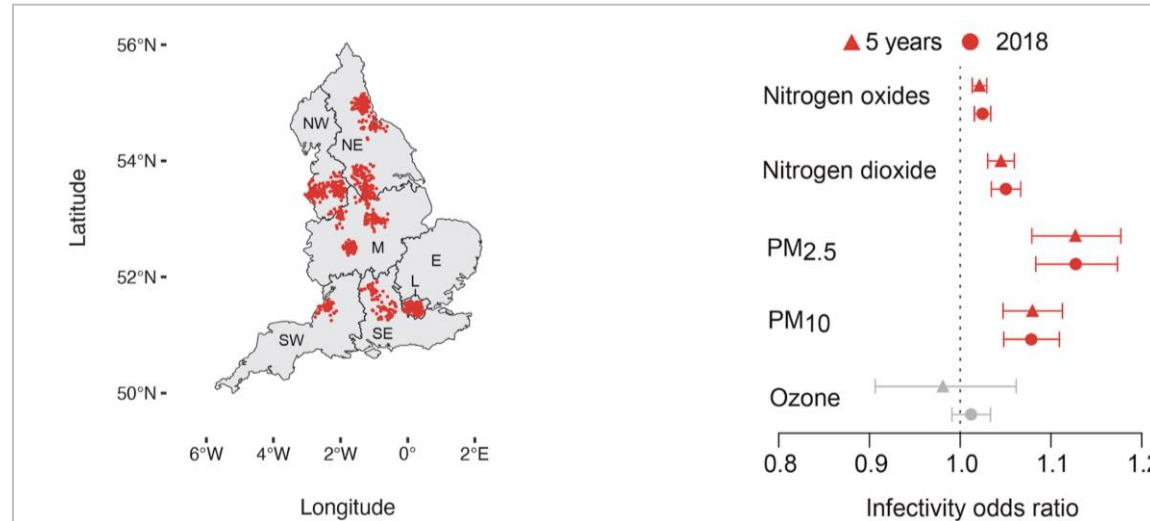
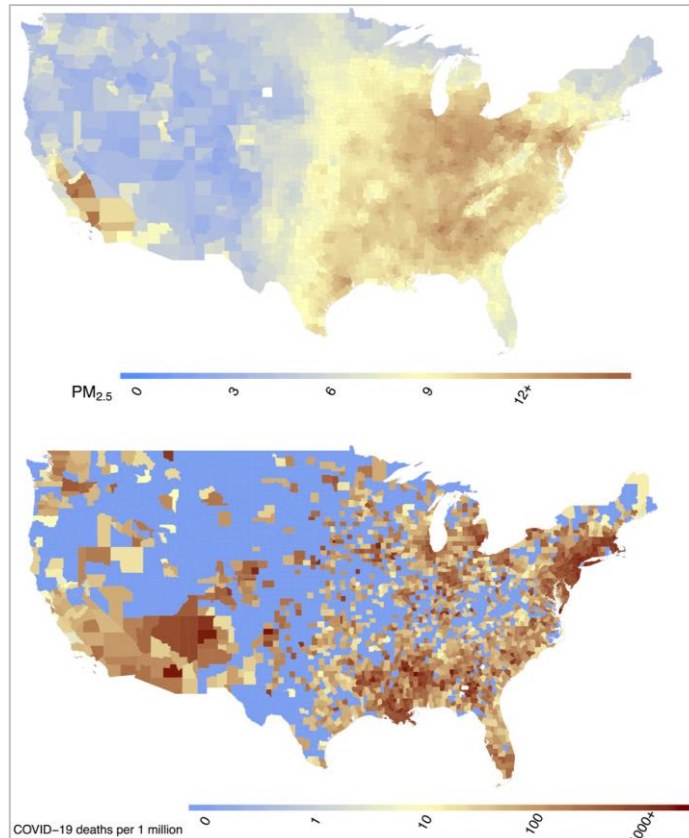
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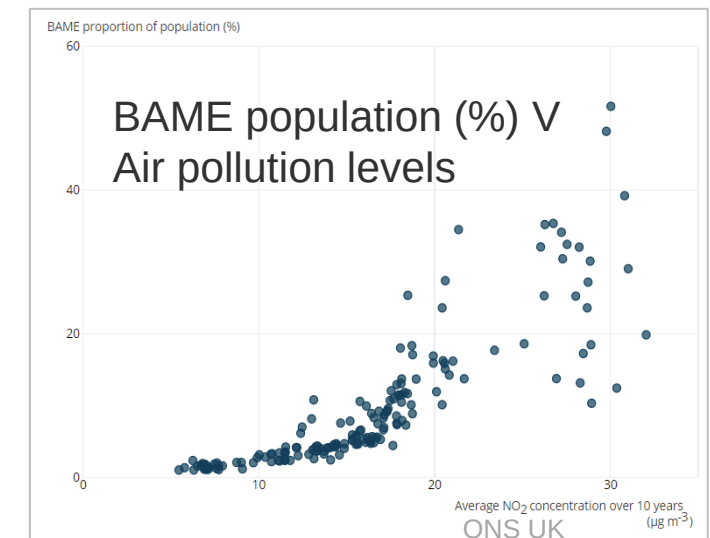
3.8 million

deaths every year as a result of exposure to indoor smoke from cooking fuels

4.2 million

deaths every year as a result of exposure to fine particulate matter

www.who.int/data



...and More

Effects of environment on mental health

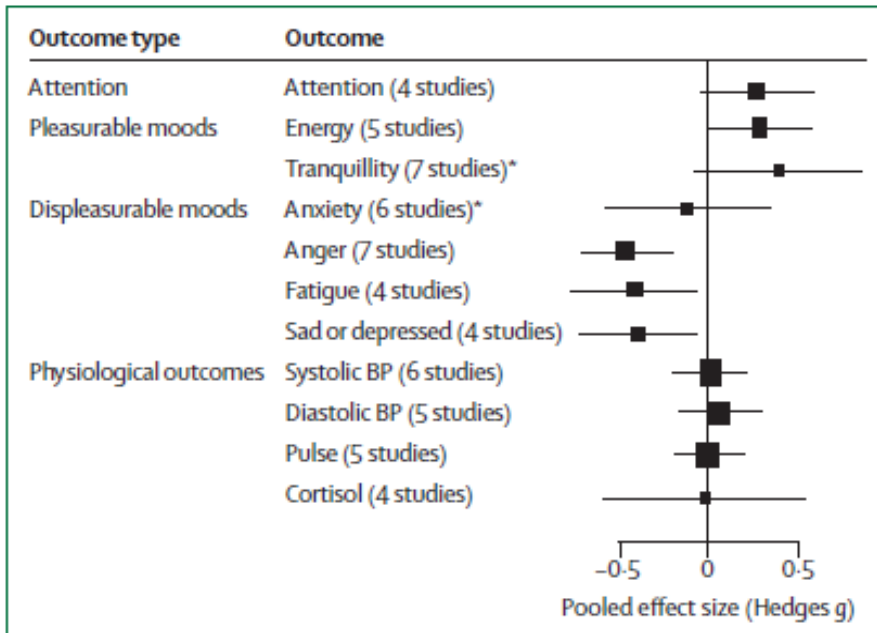
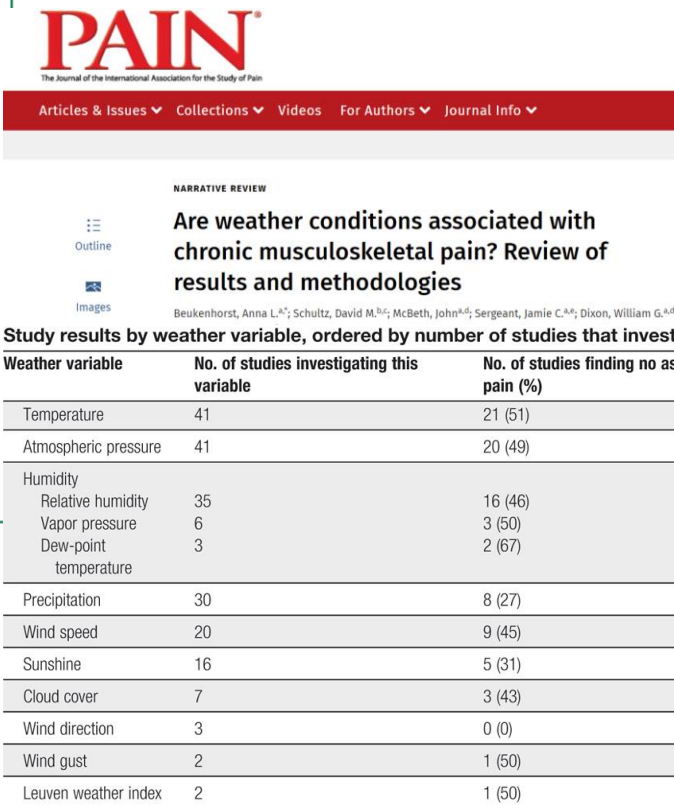


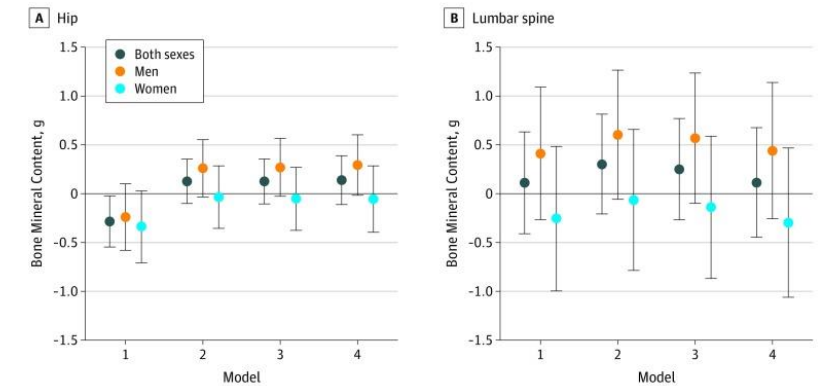
Figure 18: Psychological and emotional outcomes from exposure to natural versus synthetic environments

Bowler, D et al. BMC Public Health, 2010.
<https://doi.org/10.1186/1471-2458-10-456>

Effect of weather on pain



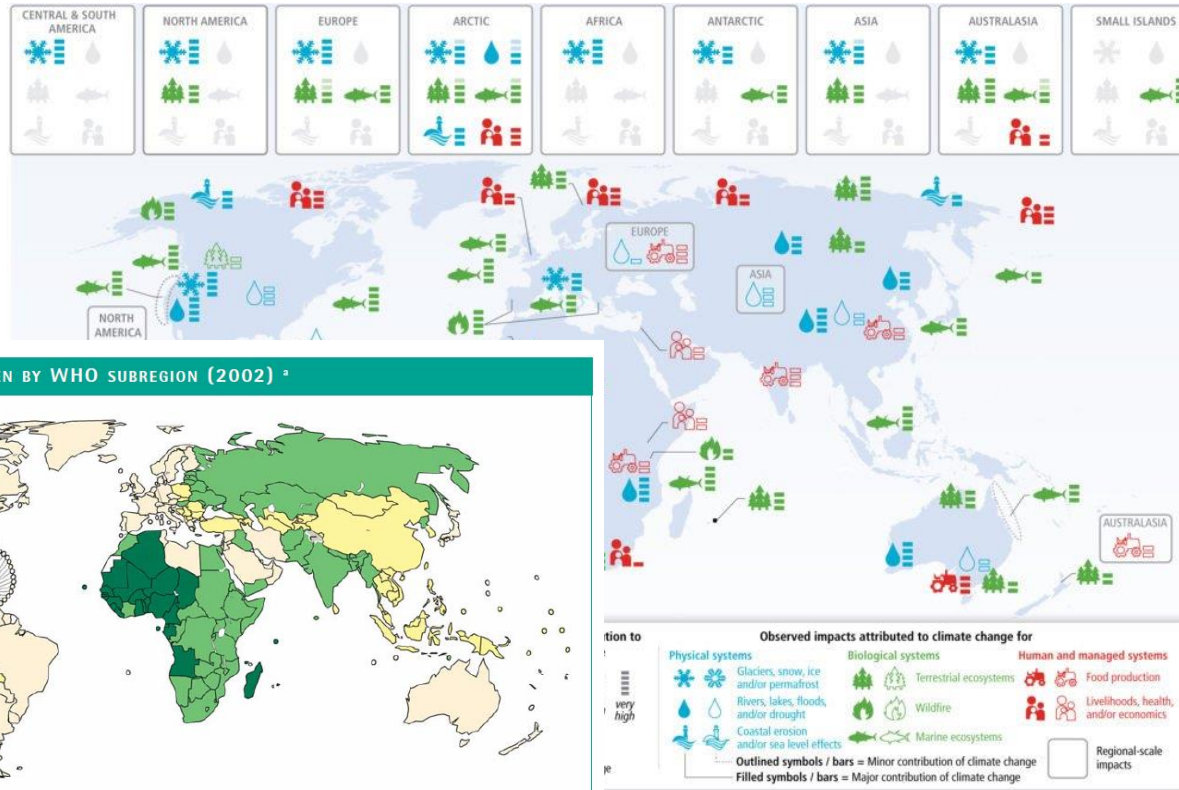
Effects of air pollution on musculoskeletal health



Ranzani, O et al. JAMA Open 2020. doi: 10.1001/jamanetworkopen.2019.18504

A Planet-wide Problem

“This is the first time a global generation of children will grow up in a world made far more dangerous and uncertain as a result of a changing climate and degraded environment.”
UNICEF



Prüss-Üstün et al. WHO Preventing Disease Report 2006

Cramer et al. Detection and attribution of observed impacts.

Climate Change 2014, IPCC 5th AR

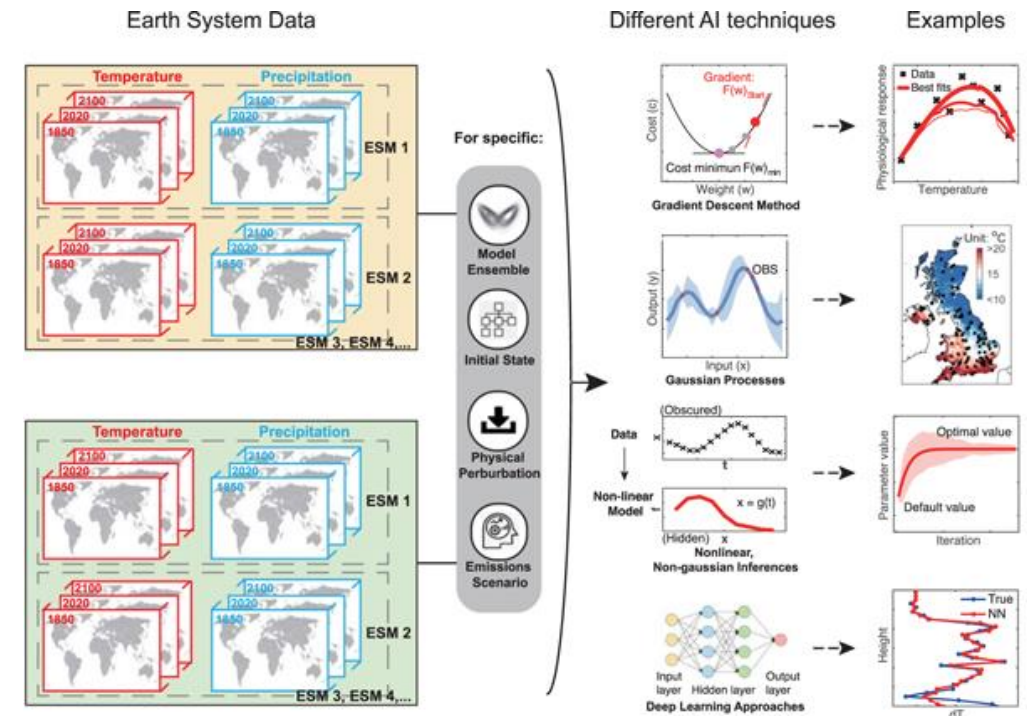
Planetary Health Informatics Group | sara.khalid@ndorms.ox.ac.uk

Centre for Statistics in Medicine | University of Oxford | @CSM_Oxford | @PHI_Oxford

Health Informatics & Planetary Health

Climate Data Science

- Earth systems modelling
 - e.g. classification of land cover
- Future climate scenarios
- Weather forecasting
- Extreme event prediction
 - e.g. flood prediction
- Climate impacts
 - e.g. on agriculture, health, economics
- Data curation and data quality
 - e.g. point-, grid-, satellite-based precipitation, temperature, vegetation, built environment data

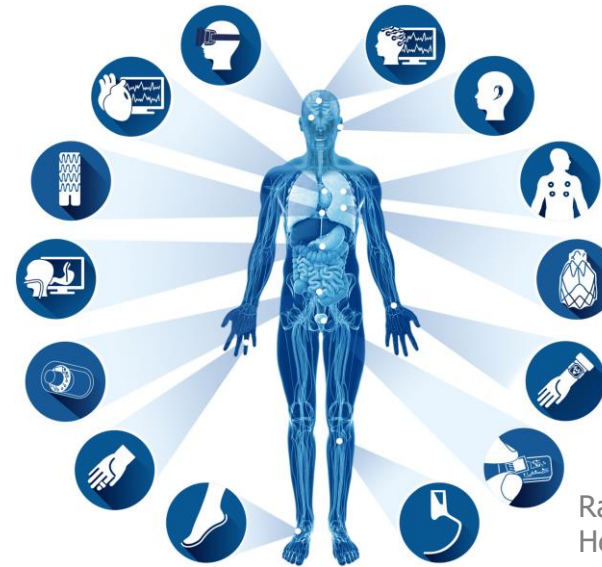


ML and AI-based understanding enhances next generation ESMs

Huntingford C et al 2019 *Environ. Res. Lett.* 14 124007

Health Data Science

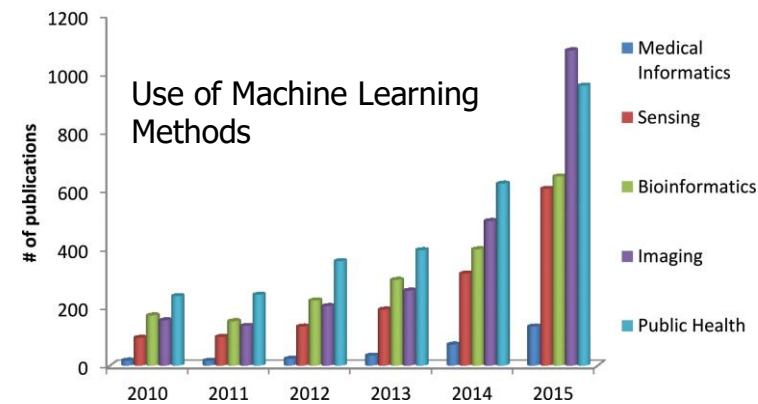
- Providing real-time, real-world **data**



Ravi et al., IEEE Biomed & Health Informatics 2017

Health Data Science

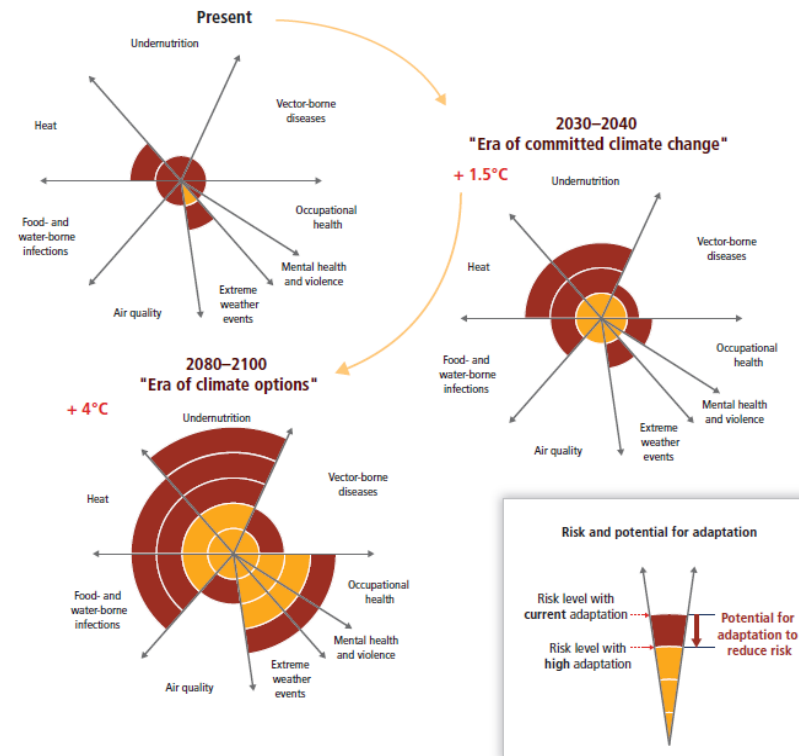
- Providing real-time, real-world **data**
- Improving **knowledge** on climate and health



Ravi et al. IEEE Biomed & Health Informatics 2017

Health Data Science

- Providing real-time, real-world **data**
- Improving **knowledge** on climate and health
- **Adaptation** of health systems to changing climate



Climate impact on 8 health sectors and potential for mitigation through adaptation.

Smith et al. Human health: impacts, adaptation, and co-benefits.

Climate Change IPCC 5th AR 2014

Early Warning Systems

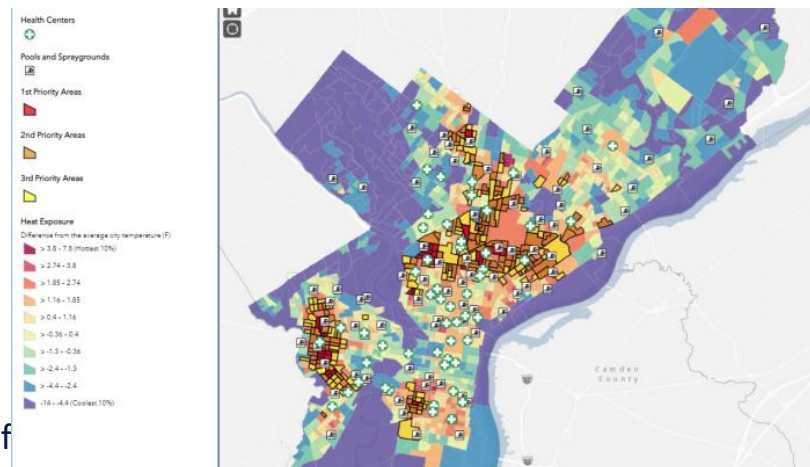
- Traditionally based on local knowledge
- Intelligent EWS based on
 - Expert knowledge
 - Real-time big data
 - Advanced analytics

Evaluation of heat-health warning systems in Shanghai, China

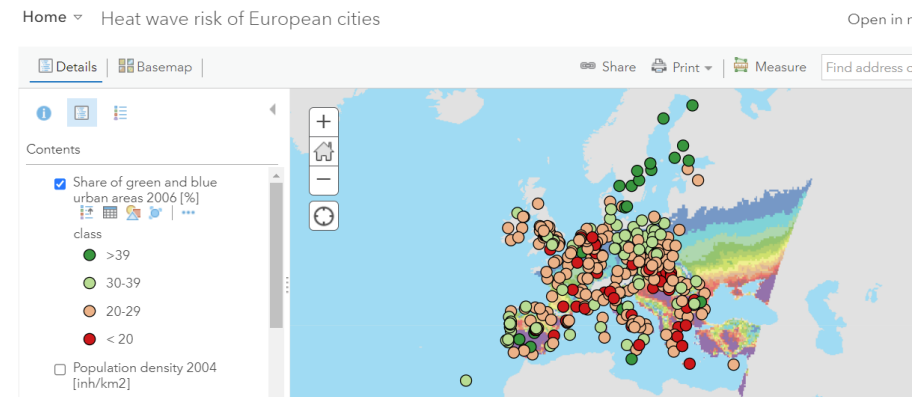


Yaqiao et al. Science of The Total Env.
<https://doi.org/10.1016/j.scitotenv.2020.136883>

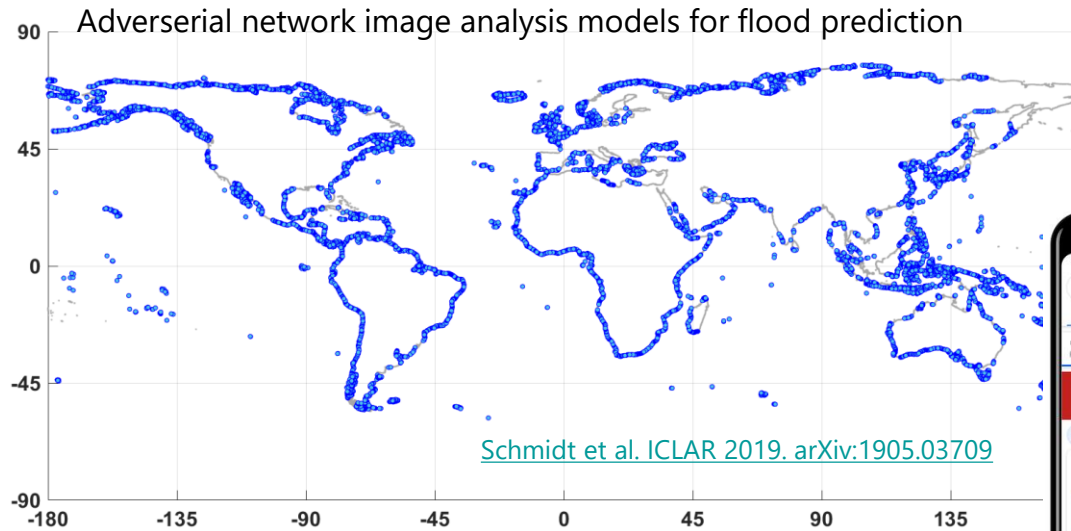
Heat risk index Philadelphia, USA.



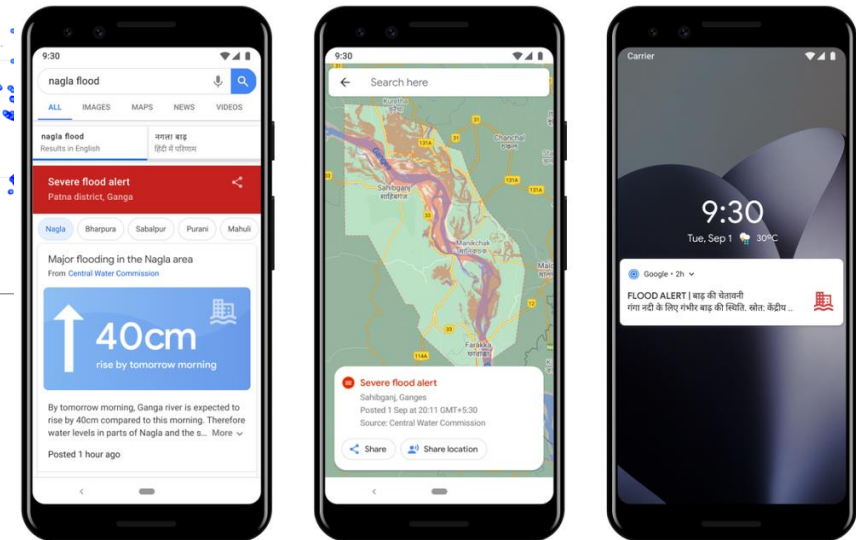
Heat risk map. European Environment Agency



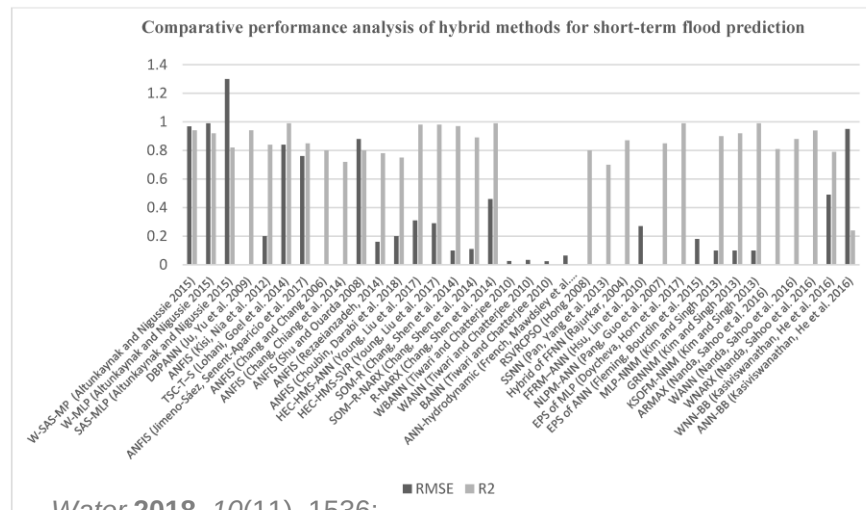
Flood Prediction Models



Google Flood Forecasting Initiative in use in India and Bangladesh



Credits: Alerts for flood forecasting, Google



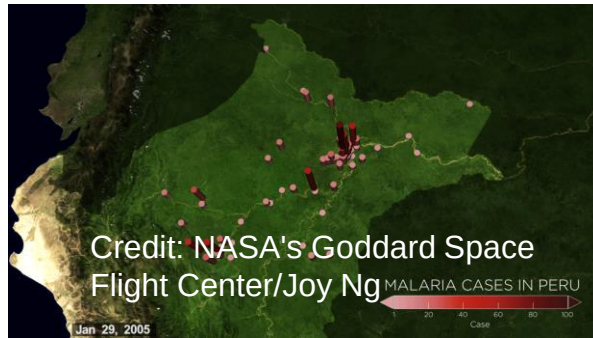
Water 2018, 10(11), 1536;

<https://doi.org/10.3390/w10111536>

Predicting Disease Outbreaks



Satellite data from Landsat series combined with ground data on precipitation, temperature, soil moisture, vegetation



Computational models tracking human and environmental events

Predicting malaria outbreaks in Peru at household level

Schmidt et al. ICLR 2019. arXiv:190

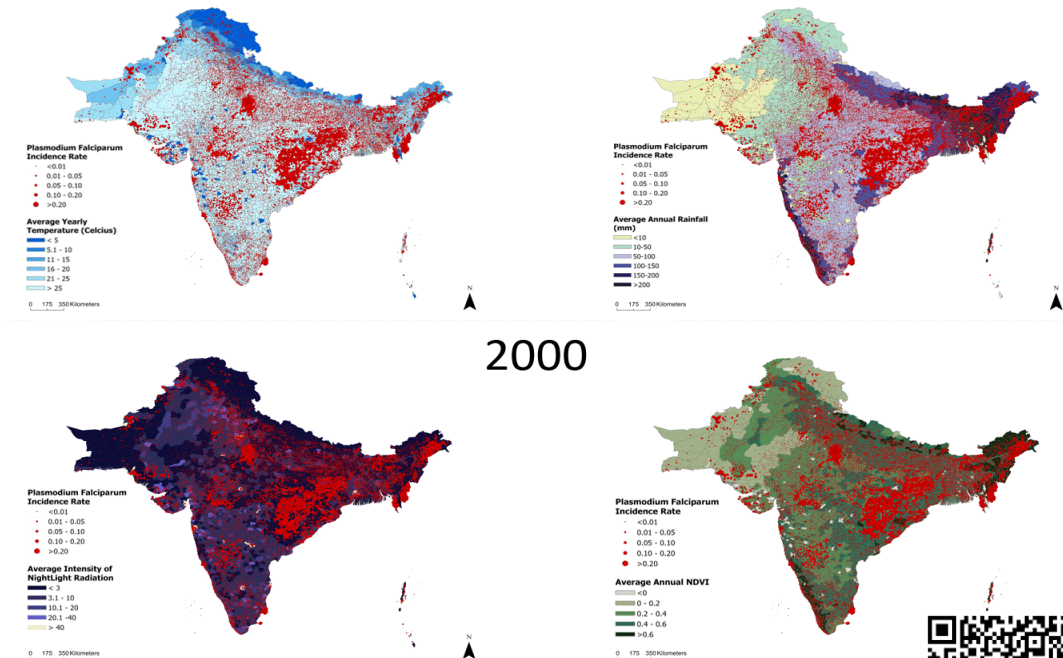


PREDICTING MALARIA OUTBREAKS USING EARTH OBSERVATION MEASUREMENTS AND SPATIO-TEMPORAL DEEP LEARNING MODELLING: A SOUTH ASIA CASE STUDY FROM 2000 TO 2017

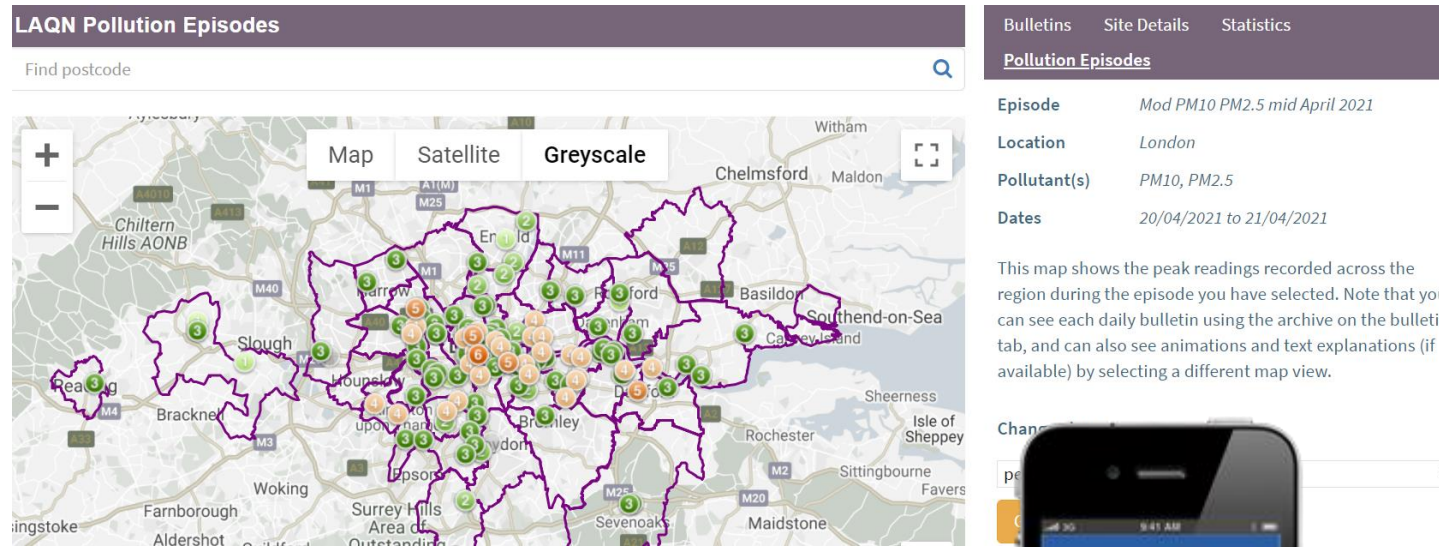
Usman Nazir¹, Muhammad Talha Quddoes¹, Momin Uppal¹, Sara Khalid²

¹Lahore University of Management Sciences, ²Centre for Statistics in Medicine, University of Oxford

sara.khalid@ndorms.ox.ac.uk



Tracking and Monitoring Air Pollution



The London Air Quality Network monitors and predicts level of ozone, nitrogen oxide, PM2.5 and PM10



Personalised early warning system for those most vulnerable to high levels of pollution

Credit: londonair.org.uk

AI and Biodiversity

IMAGE

BROWSE... No file selected

Detection threshold:

0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1

0.68

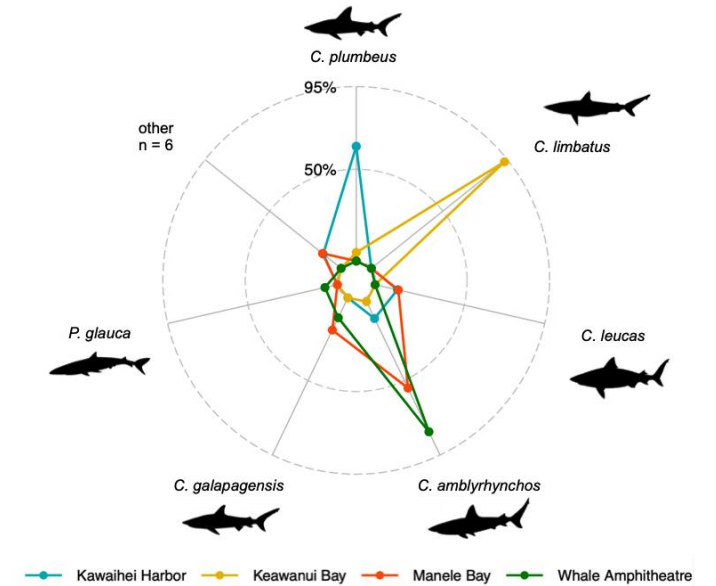
Select Video

20220924_KeawanuiBay_S3_GX110006.mp4

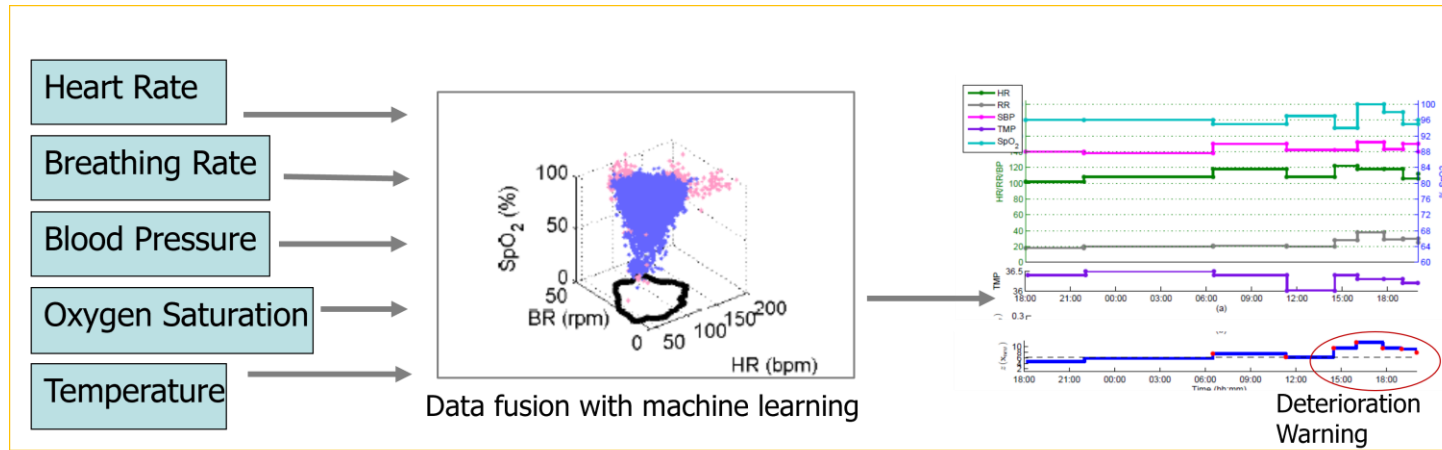
PROCESS DETECT VIDEO PLAY VIDEO

RESET

Shark	Prediction
Carcharhinus limbatus	94.15 %
Carcharhinus plumbeus	4.13 %
Carcharhinus amblyrhynchos	0.47 %
Galeocerdo cuvier	0.44 %
Carcharodon carcharias	0.39 %



Remote Monitoring of Patients in Disaster/Conflict Zones



Home monitoring kit for remote management of Covid-19 infection

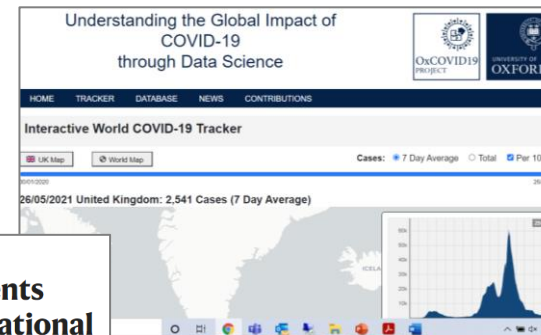
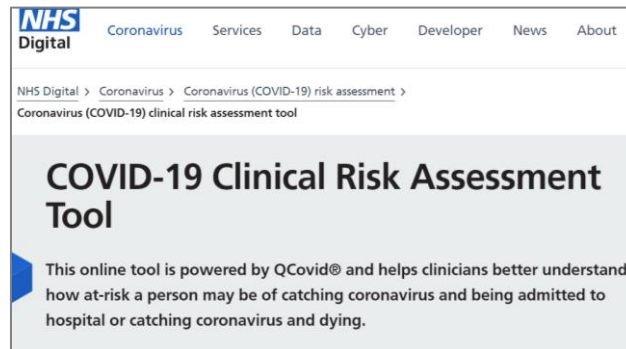
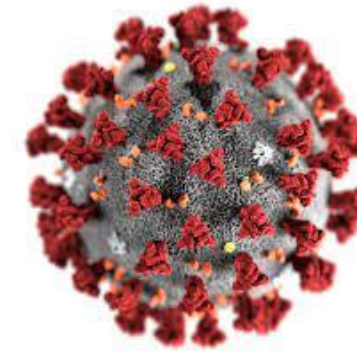


Khalid et al. IEEE Trans Biomed Eng 2012
DOI: 10.1109/TITB.2012.2212202

Santos et al. Front. Digit. Health
<https://doi.org/10.3389/fdgth.2021.630273>

Pandemic Response

- Diagnostics
- Clinical Risk Prediction
- Vaccine Strategy
- Deep Phenotyping



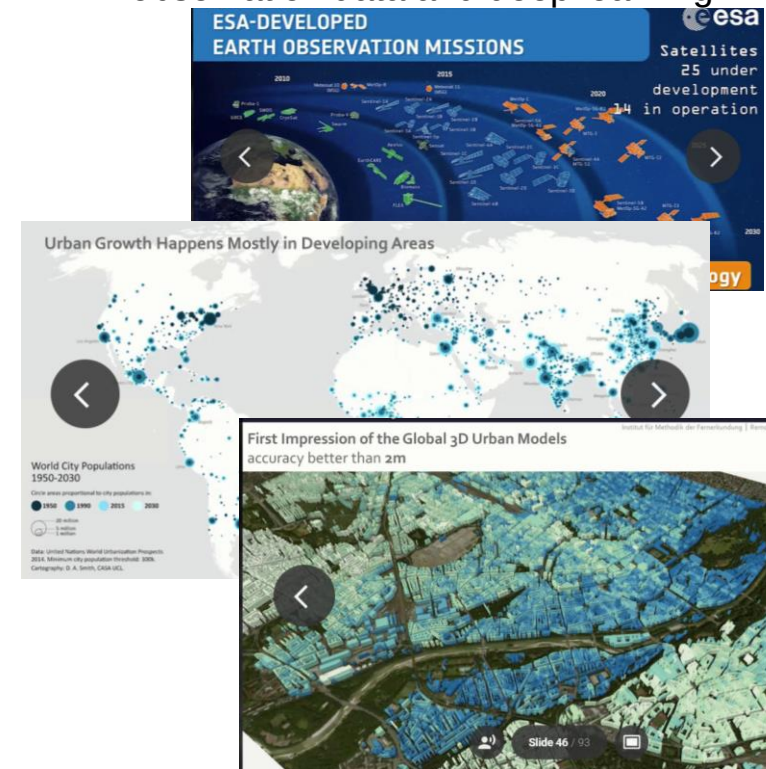
Article | Open Access | Published: 06 October 2020
Deep phenotyping of 34,128 adult patients hospitalised with COVID-19 in an international network study
 Edward Burn, Seng Chan You, [...]Patrick Ryan
Nature Communications 11, Article number: 5009 (2020) | Cite this article

HI_Oxford

Computational Models

- Future cities
 - Urban thermal comfort
 - Land use optimisation
 - Increased vegetation cover
- Transport modelling
 - More physical activity
 - Less carbon footprint
- AI for renewable energy
- Carbon footprint of AI

Building a 3D blueprint of urban hot spots using earth observation data and deep learning



XX Zhu et al. IEEE Trans. Geoscience & Remote Sens 2017

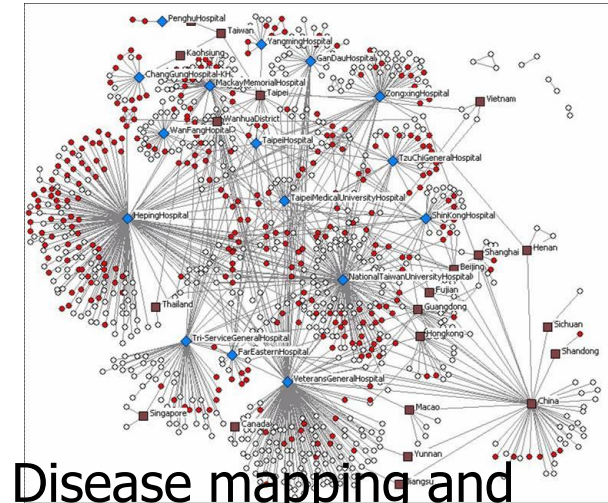
Real-world Applications



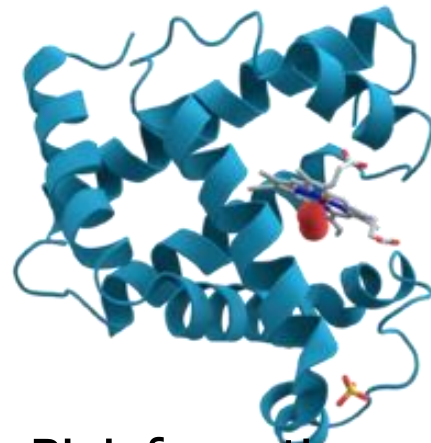
Eye disease detection



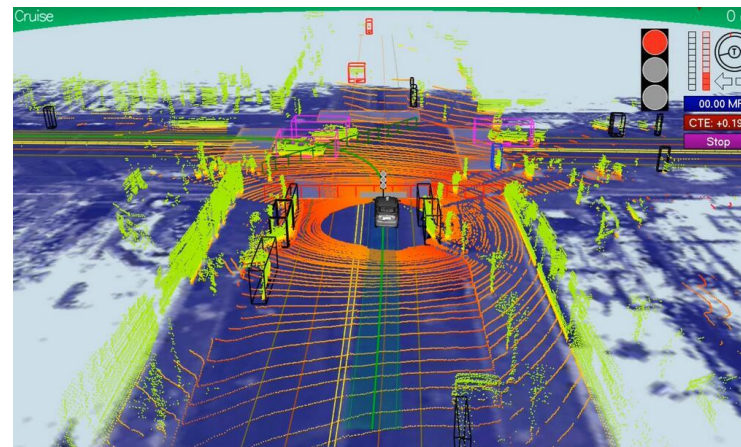
Tumour diagnosis



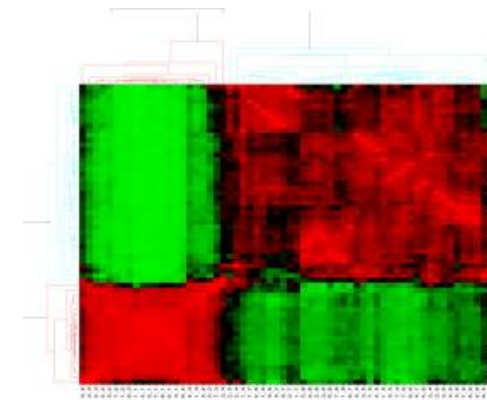
Disease mapping and tracking



Bioinformatics



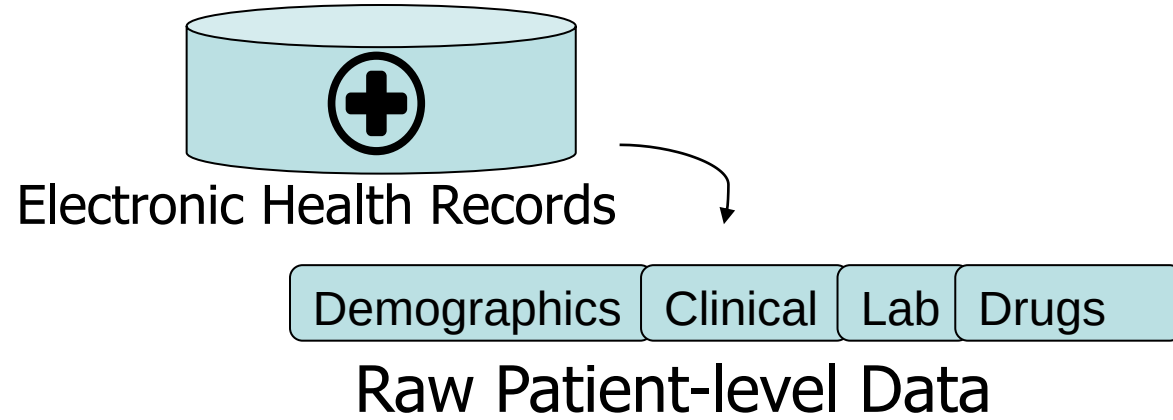
Autonomous vehicle navigation



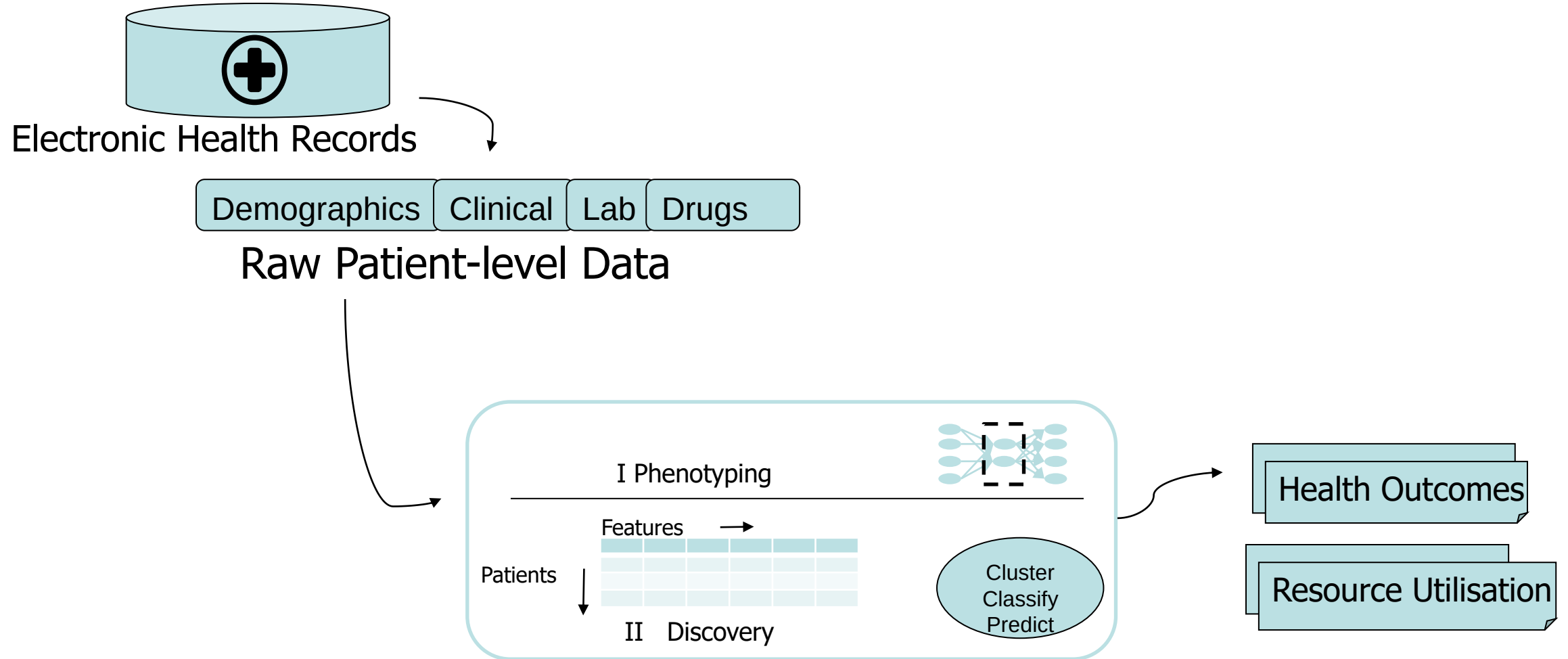
Gene expression clustering

Challenges and Opportunities

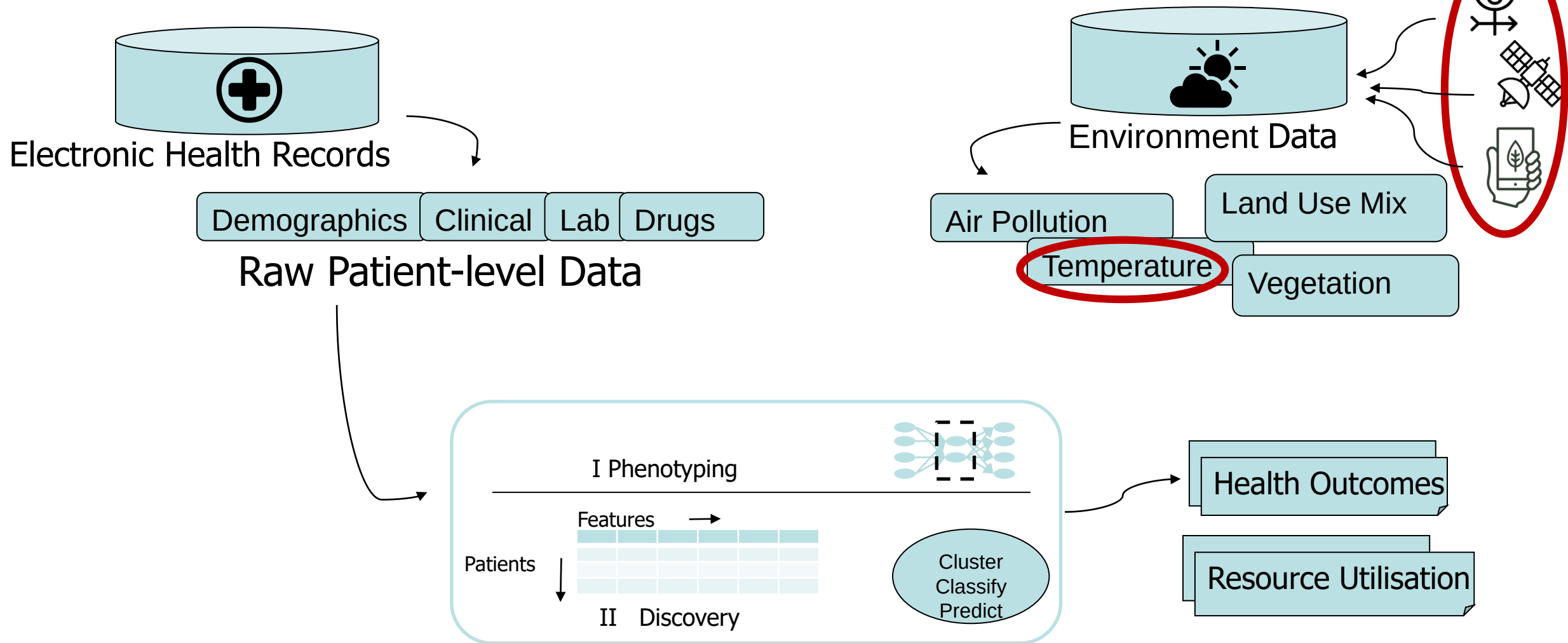
Planetary Health Analytics Framework



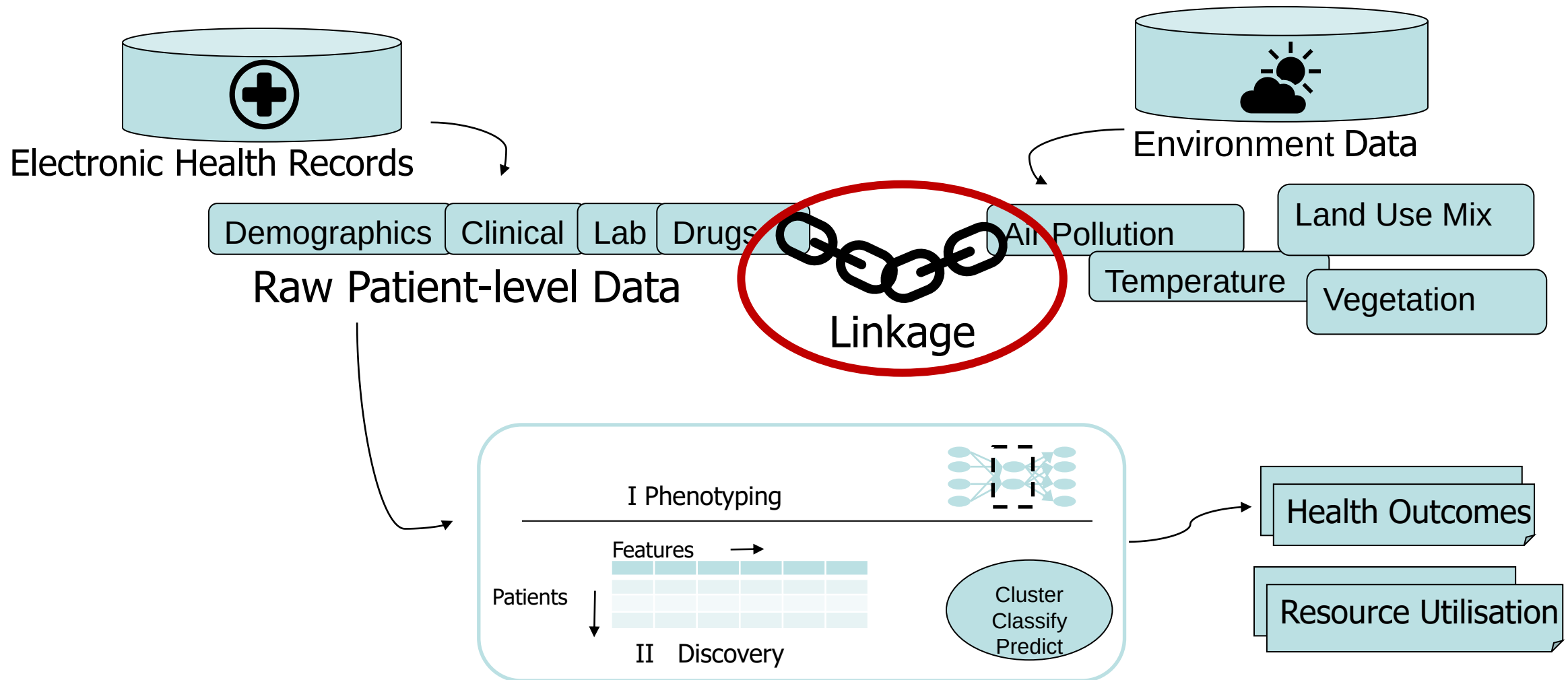
Planetary Health Analytics Framework



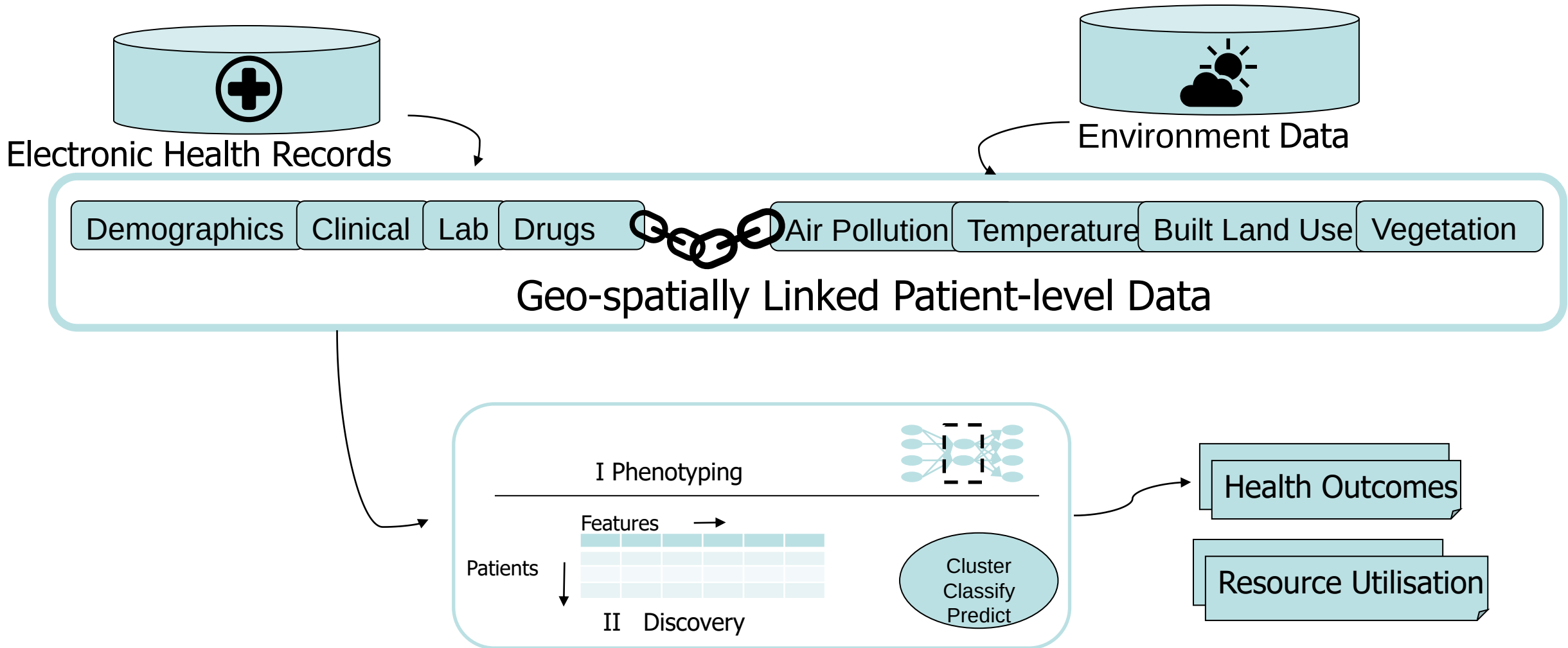
Planetary Health Analytics Framework



Planetary Health Analytics Framework

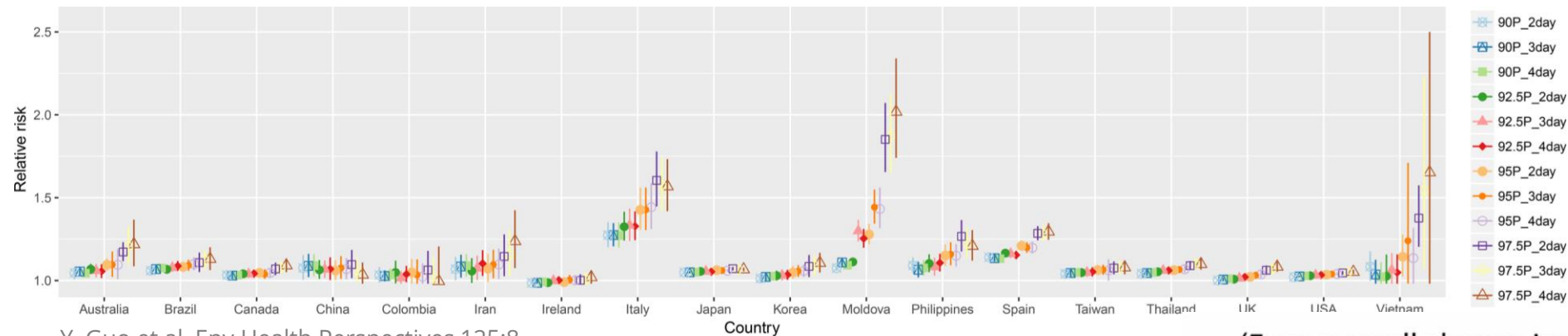


Planetary Health Analytics Framework



Heat Definition, Coding, and Capture

Effect of heat wave on mortality in 18 countries with 12 different definitions of heat wave



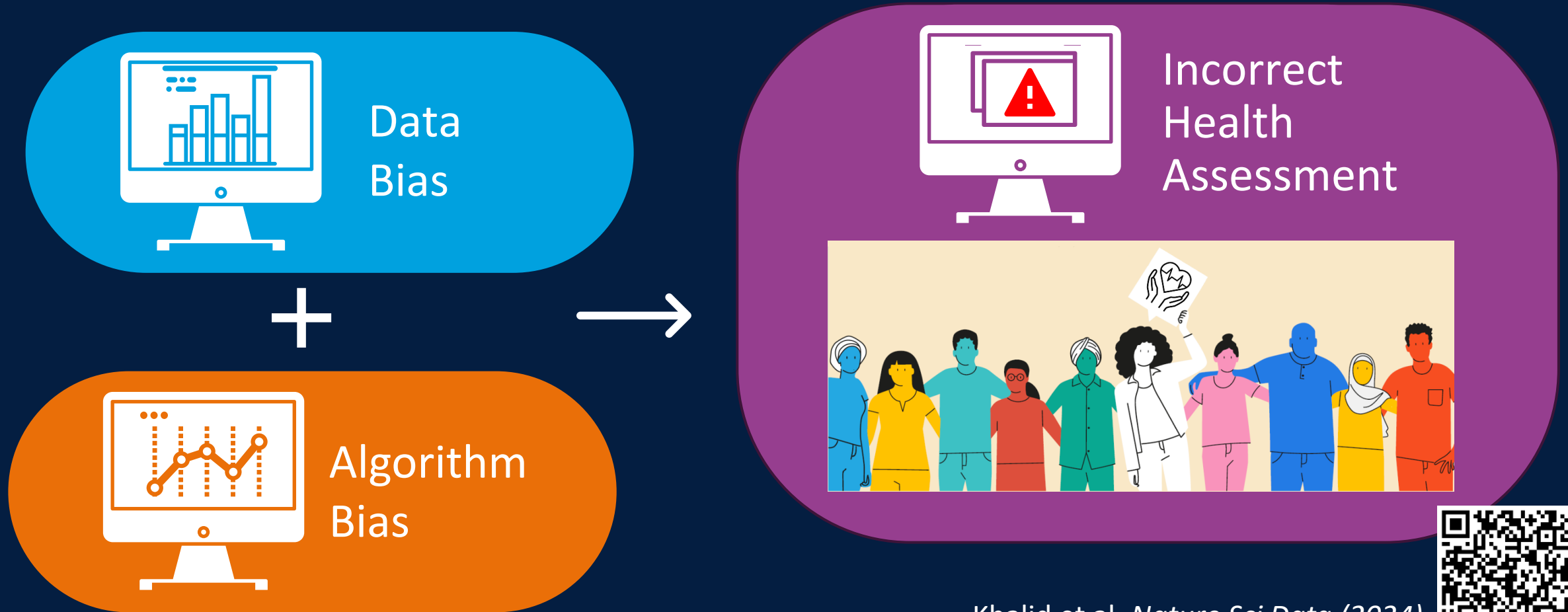
Y. Guo et al. Env Health Perspectives 125:8
CID: 087006 <https://doi.org/10.1289/EHP1026>

‘Even a small change in the heatwave definition had an appreciable effect on the estimated health impact’ in Brisbane, Australia - Case Cross Over Design

HWD	Definition	Mortality	OR with 95% CI
1	The daily maximum temperature $\geq 35^{\circ}\text{C}$ (about top 1%) for 3 or more consecutive days	HWD1	0.87 0.68, 1.10
2	The daily maximum temperature of more than 5 consecutive days exceeds the average maximum temperature by 5°C , the normal period being 1961–1990	HWD2	0.99 0.83, 1.18
3	The heat index (maximum temperature + relative humidity) is expected to reach 40.6°C with a minimum temperature not below 26.7°C as a period of at least 48 h	HWD3	1.73 1.32, 2.28
4	The daily maximum temperature would be equal to or greater than 35°C (about top 1%) for at least consecutive 2 days	HWD4	1.26 1.10, 1.43
5	The daily maximum temperature would be equal to or greater than 37°C (about top 0.5%) for at least consecutive 2 days	HWD5	1.53 1.27, 1.84
6	The top 2.5% ($\geq 33.59^{\circ}\text{C}$) of daily maximum temperatures for a continuous 2 days period	HWD6	1.16 1.06, 1.27
7	The top 2.5% ($\geq 33.59^{\circ}\text{C}$) of daily maximum temperatures for a continuous 3 days period	HWD 7	1.21 1.08, 1.35
8	The top 5% ($\geq 32.65^{\circ}\text{C}$) of daily maximum temperatures for a continuous 3 days period	HWD8	1.10 1.03, 1.18
9	The top 5% ($\geq 32.65^{\circ}\text{C}$) of daily maximum temperatures for a continuous 4 days period	HWD9	1.14 1.05, 1.24
10	The top 5% ($\geq 32.65^{\circ}\text{C}$) of daily maximum temperatures for a continuous 5 days period	HWD10	1.24 1.13, 1.36

Tong et al. 2010

Real-world Data, AI & Health Equity



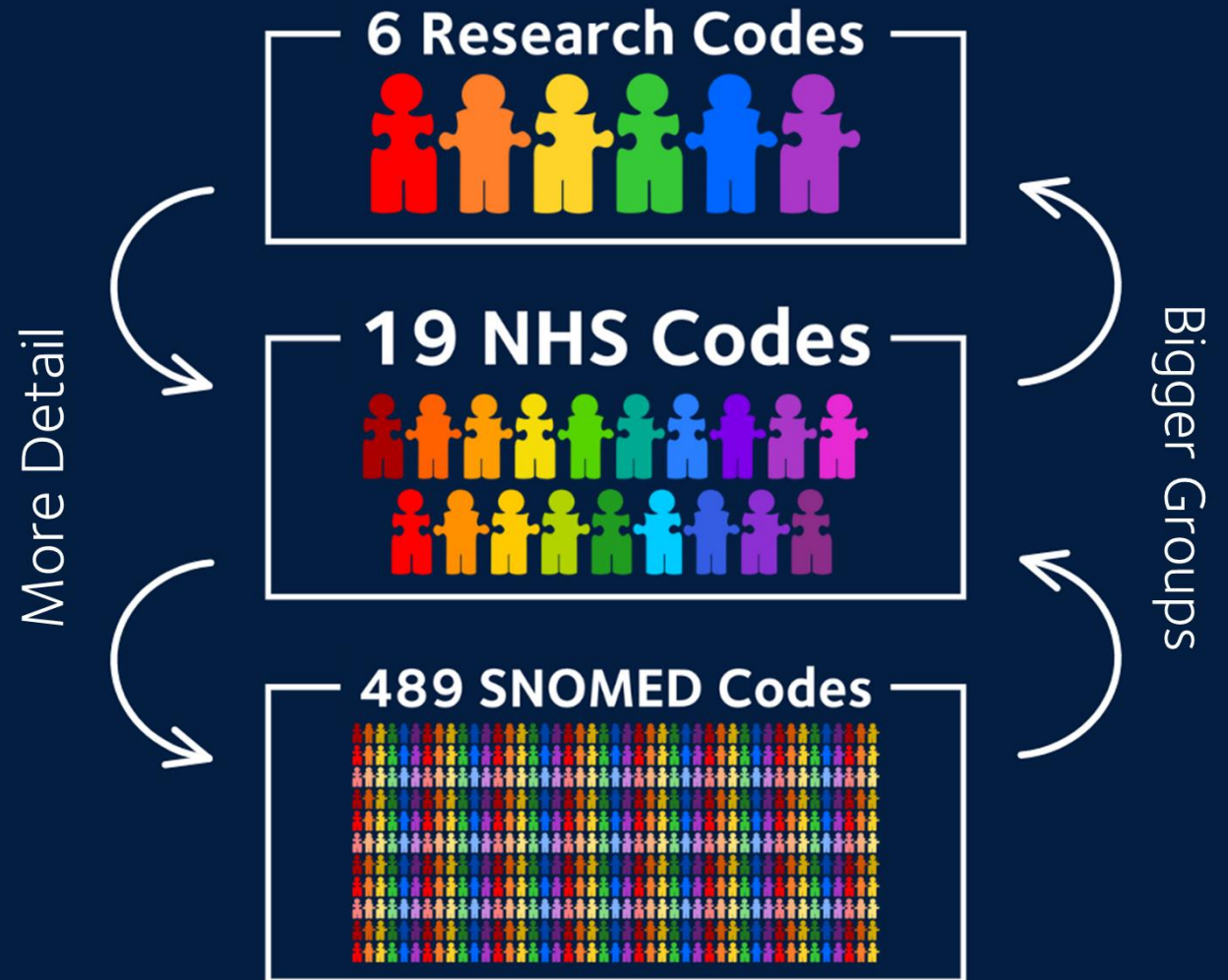
Khalid et al. *Nature Sci Data* (2024)



Real-World Data, AI & Health Equity

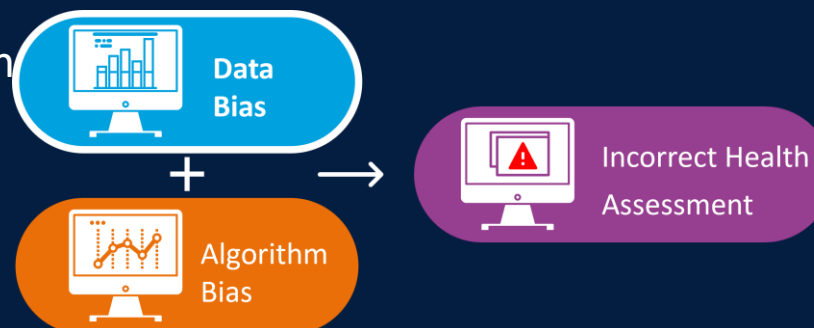
60 Million
de-identified patient records

What Ethnicity
Data is available?



Ethnicity, Health Equity and AI Project

A UK Government COVID-19 Core Study aimed at addressing biases in data which translated to health inequalities. Analysing data from NHS England revealed previously unused, rich ethnicity data.



60m

Patients,
representing **94%**
of UK
population

255

Patient self-identified,
using the SNOMED system

1/10

Patients have missing
ethnicity data across their
primary and secondary care
records

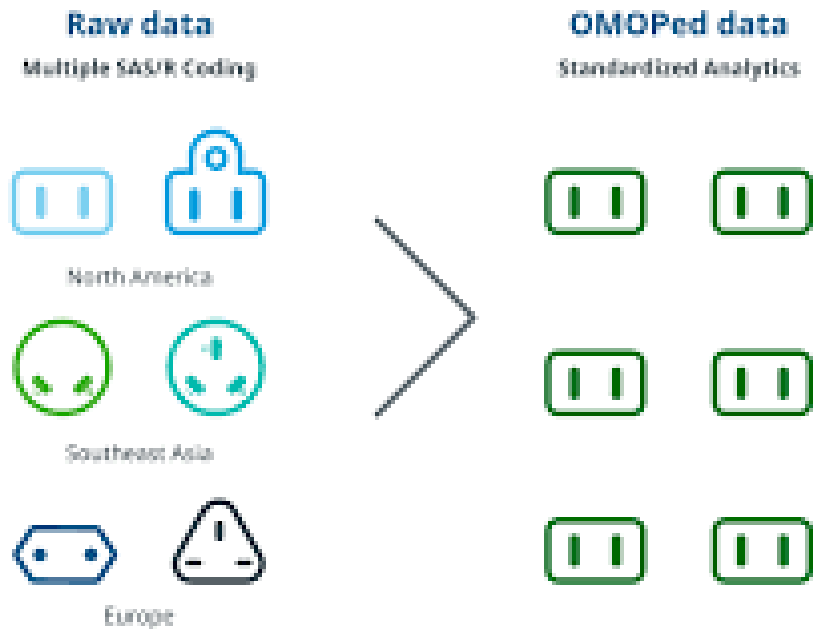
17%

Of patients have
conflicting ethnicity codes
in their health records

Common Data Models & Standardised Analytics

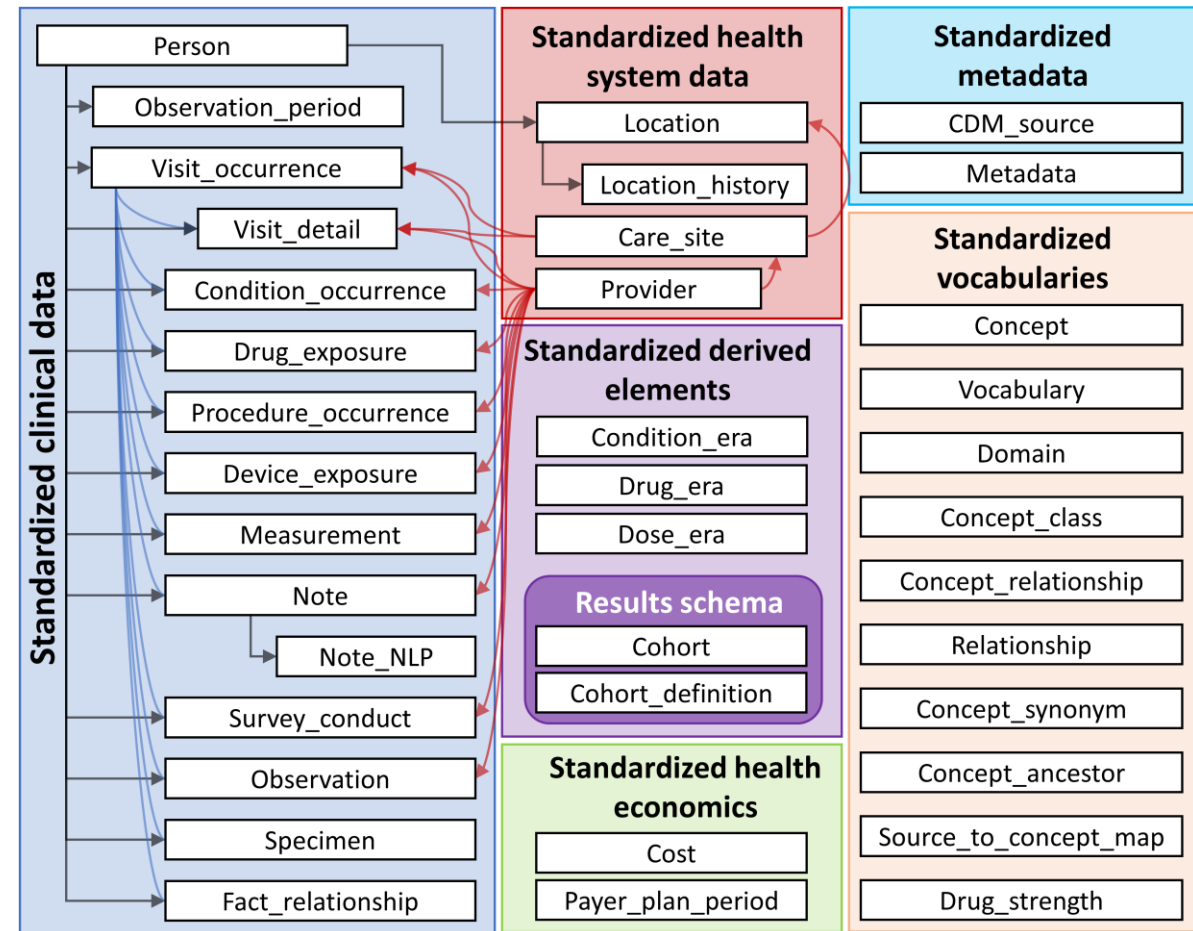
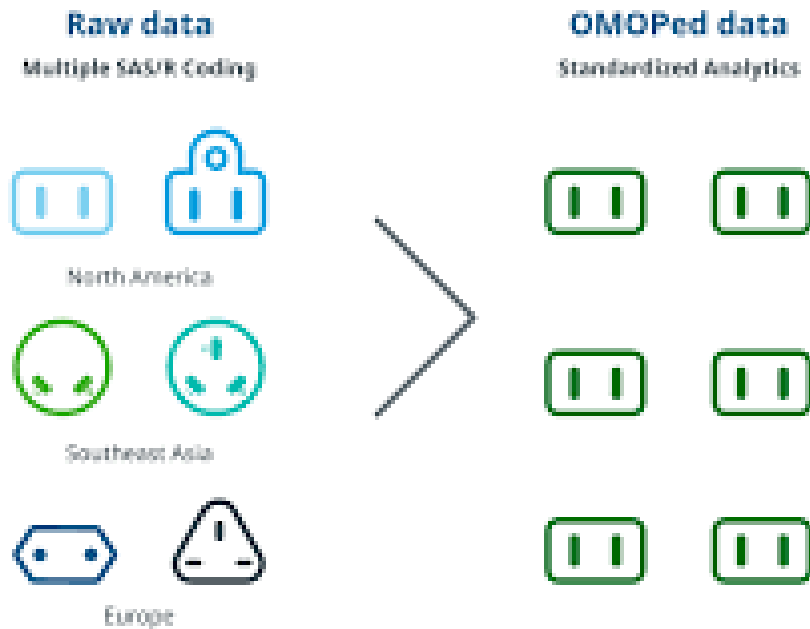


Common Data Models & Standardised Analytics



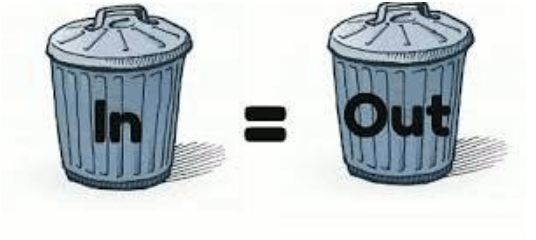
Common Data Models & Standardised Analytics

Example of a common data model developed by OHDSI

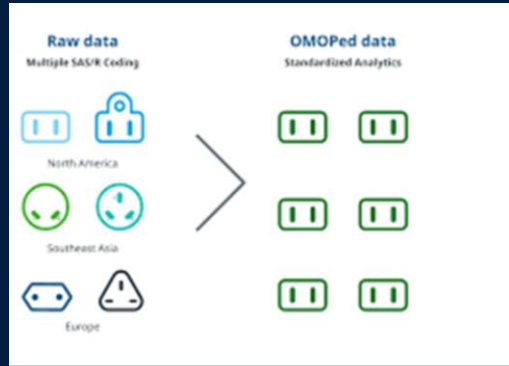


Considerations

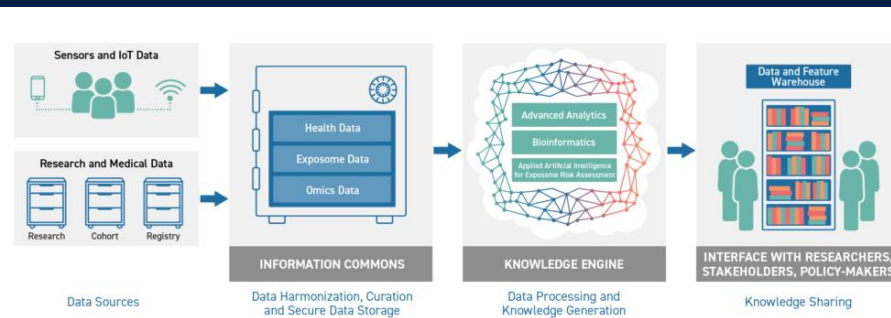
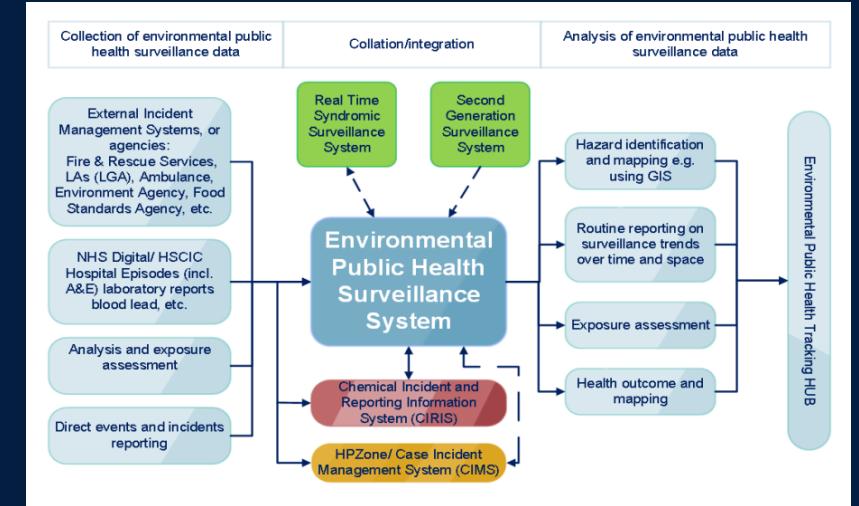
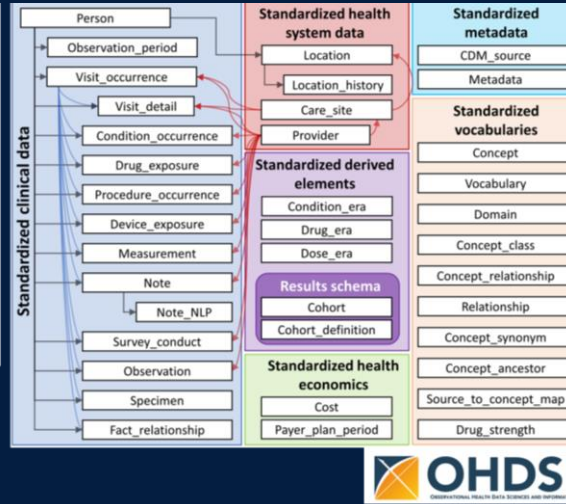
- Data Sources
 - Global, ethical, representative
- Linkage
 - De-identified data
- Standardisation
 - Universal definitions
 - Temporal synchronization
 - Common data model
- Data Quality
 - Completeness and accuracy of recording (e.g. heat)
- Data Analysis
 - Many unknowns
 - Confounding
 - Multi-stressor interactions



Infrastructure for Large-scale Planetary Health Analytics



Common data models and standardised analytics



HEAP Europe An AI-powered platform for research on climate impact on health

human exposome assessment platform

A project receiving funding from the European Union's Horizon 2020 Research and Innovation programme under Grant Agreement N. 101004188

DEEP CUBE

DeepCube
Explainable AI pipelines for big Copernicus data



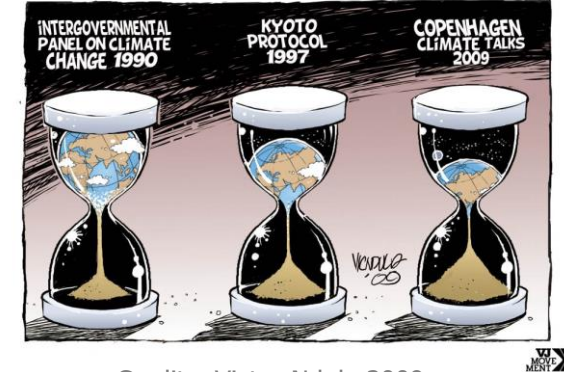
Deep Cube Using AI to analyse Copernicus satellite data and predict climate impacts

Public Health England

EPHSS Environmental public health surveillance system UK

Climate-health Co-benefits

- Global health adaptation **spending** \$18Bn, ↑ 5.3%
- Health systems adaptation **planning** in 86 countries
- Ambitious **net-zero** healthcare plans, e.g. UK NHS care delivery emissions ↓ 57%
- Health providers **divestment**: \$42Bn
- **Research** on health and climate is up 8 times
- Economic **benefit** of 1.5C: \$600 trillion
 - healthier workforce, reduced healthcare costs



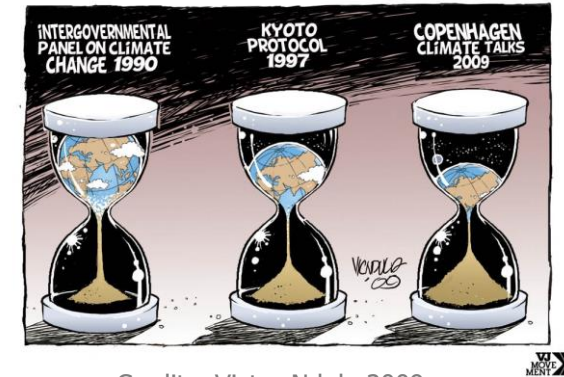
Credits: Victor Ndula 2009

Lancet Countdown Report 2020

Climate-health Co-benefits

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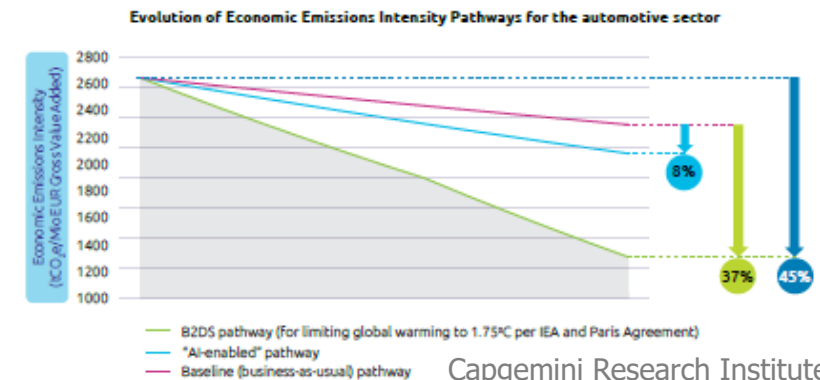
Lancet Countdown Report 2020



Credits: Victor Ndula 2009

Forbes AI to help meet 45% Paris 2015 targets!

Figure 6. AI-enabled use cases have the potential to assist automotive organizations to meet more than one-fifth of their Paris Agreement emission goals by 2030



Handle with Care

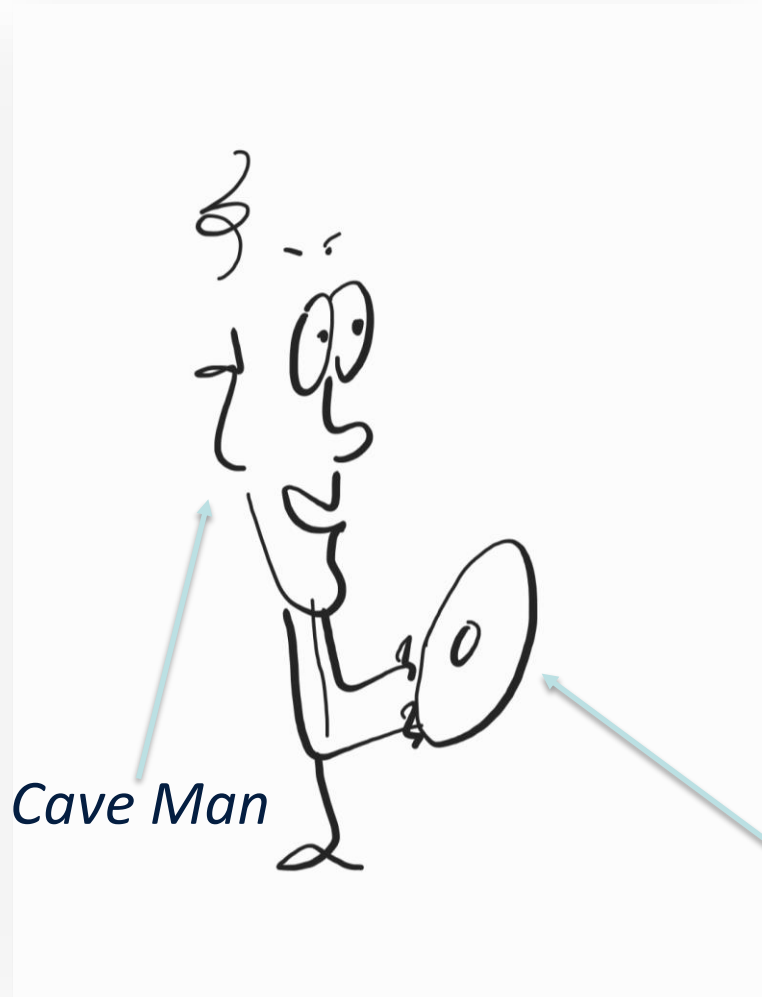
Top scientists call for caution over artificial intelligence

Artificial intelligence has the potential to eradicate disease and poverty, say world's top scientists, but researchers must not create something which cannot be controlled



The Telegraph (2015)

There is a choice



The “first” wheel

The first wheel (not a donut!)

The choice is ours

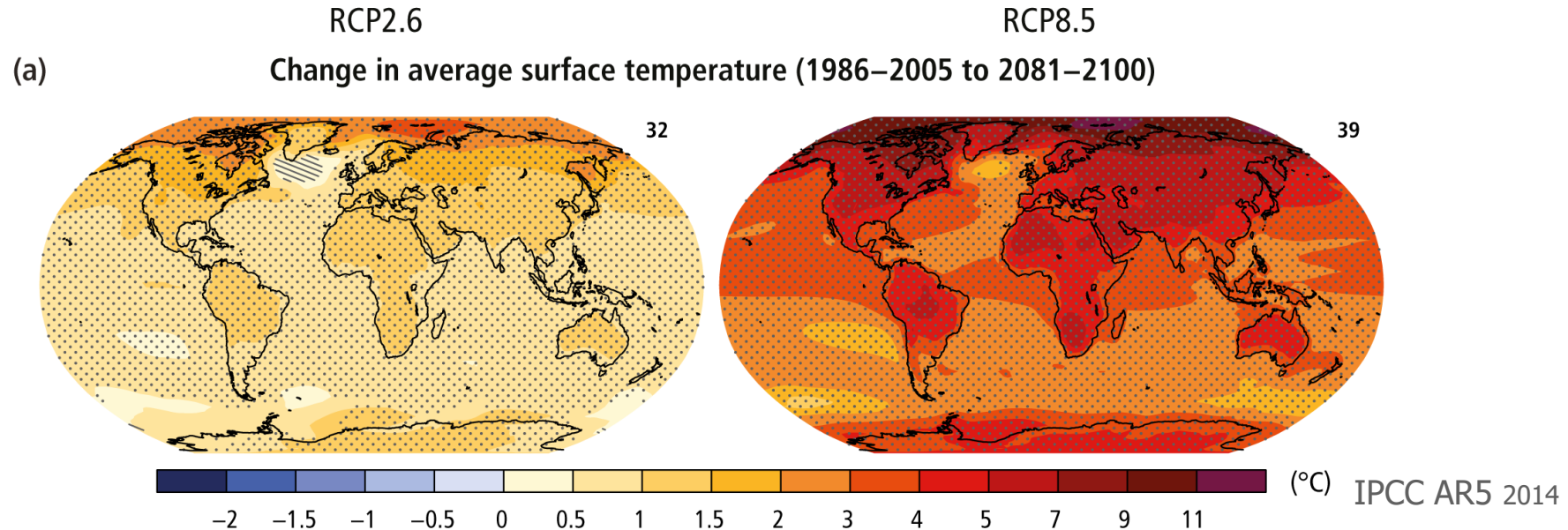


The first wheel (not a donut!)

The choice is ours



Our Rapidly Warming World



“Climate change is the biggest global health threat of the 21st century”

Lancet Commission Report 2009

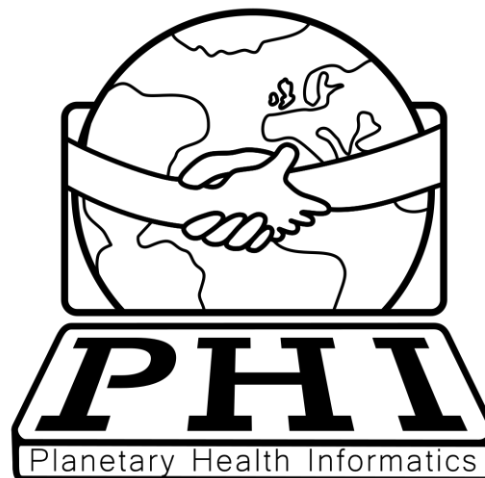
“Addressing climate change could be the biggest global health opportunity of the 21st century”

Lancet Commission Report 2015



Φ-lab

BILL & MELINDA GATES foundation





Artificial Intelligence for Planetary Health

Challenges, Opportunities, and Human Choice

The Climate Change AI Summer School 2024: Climate & Health Series

Sara Khalid *BE MSc DPhil Oxon*

Associate Professor | Centre for Statistics in Medicine

Head | Planetary Health Informatics Lab

University of Oxford

sara.khalid@ndorms.ox.ac.uk

www.csm.ox.ac.uk | [@PHI_Oxford](https://twitter.com/PHI_Oxford)

www.ndorms.ox.ac.uk/planetary-health-informatics



Planetary
Health
Informatics



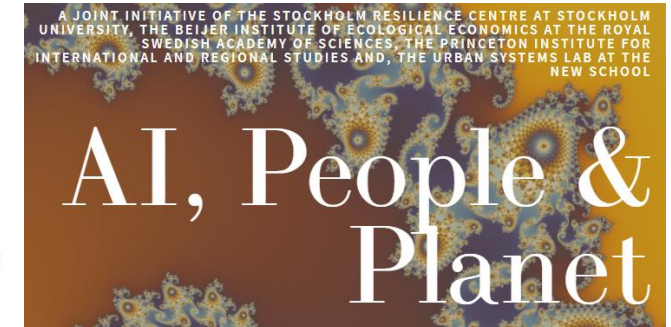
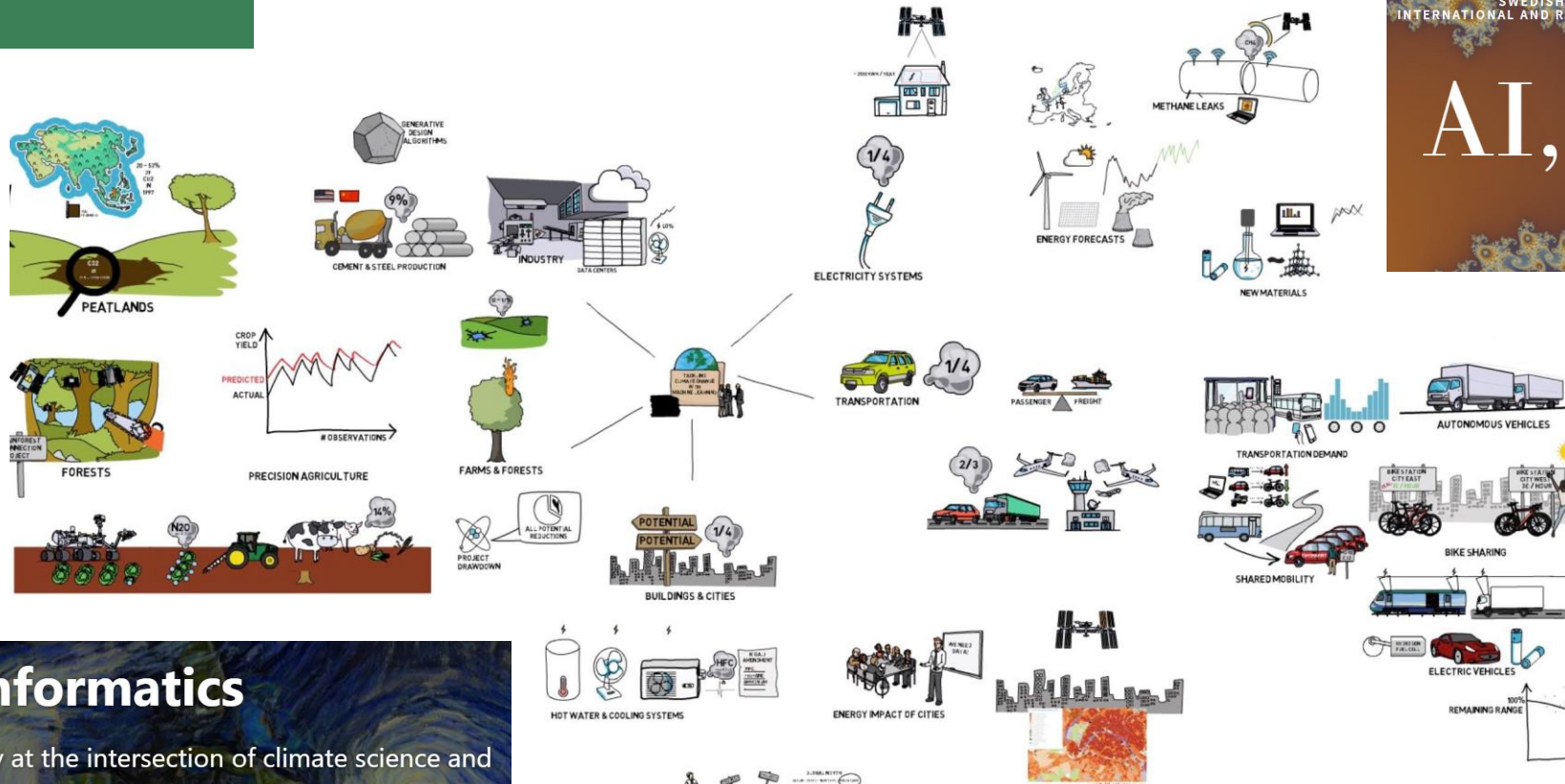
[Planetary Health Informatics Homepage](#)



@PHI_Oxford



A Call to Arms



Climate Informatics

An open community at the intersection of climate science and data science.

[Join the Community](#)

CompSustNet
Computational Sustainability Network

cor